

Formulating a Mixture of Experts with a Hidden Markov Gate

From Bishop's mixture of experts to a regime-gated cross-sectional return model

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How to use these notes

These notes build the thesis model one layer at a time, starting from Bishop’s textbook mixture of experts and ending with the full regime-gated model. They cover the *formulation* only: what the model is, which distribution it defines, what its likelihood is, and what it predicts. How we train it in the code (Adam, early stopping, walk-forward folds) comes later.

The order is deliberate.

1. **Section 1** translates Bishop’s notation into ours.
2. **Section 2 is the warm-up:** a reading guide to Chamroukhi & Nguyen (2018) and Bishop, followed by two exercises. Exercise A1 is a mixture of experts with *linear-regression* experts and a softmax gate. Exercise A2 replaces the softmax gate with a hidden Markov chain. Because the experts are linear, EM has closed-form steps, so you can carry out every derivation by hand.
3. **Sections 3 to 5** replace the linear experts with MLPs. They explain where the softmax gate comes from, why gradient training of the likelihood behaves like EM, how backpropagation works, and how the softmax gate becomes an HMM gate.
4. **Sections 6 to 9** assemble our model: the full generative model, the likelihood, prediction and posteriors, and a checklist of assumptions and open design choices.
5. **Appendices A and B** give the full solutions to the two exercises. *Attempt the exercises first.*
6. **Appendix D** gives the theory of the per-date and per-row likelihoods behind Q24.

Notation used throughout

Symbol	Meaning
$t = 1, \dots, T$	trading day
$i \in \mathcal{S}_t, N_t = \mathcal{S}_t $	stocks in the universe on day t
$x_{i,t} \in \mathbb{R}^d$	characteristics of stock i , known at the close of day t
$r_{i,t+1}$	the target : the <i>market-neutral</i> return of stock i over $(t, t+1]$, i.e. its return minus that day's equal-weighted cross-sectional mean (decided 2026-10-01, Q25)
m_t	the gate's input series: the daily log return of the CRSP value-weighted S&P 500 index (Q23)
$\mathcal{F}_t = \sigma(m_1, \dots, m_t)$	the gate's information at the close of day t : the σ -algebra generated by the market series up to t . The family $(\mathcal{F}_t)_{t \geq 1}$ is a filtration (Section 1.4)
$\mathcal{G}_t \supseteq \mathcal{F}_t$	all information at the close of t : market series, characteristics and returns up to t
$z_t \in \{1, \dots, K\}$	market regime on day t (one-hot vector z_{tk})
A_{jk}, ρ_k	transition probability $p(z_{t+1}=k \mid z_t=j)$ and initial probability $p(z_1=k)$
$\eta_t(k) = \mathcal{N}(m_t; \nu_k, \sigma_k^2)$	gate emission density: market series in regime k
$\xi_{t \tau}(k)$	$p(z_t=k \mid m_1, \dots, m_\tau)$: filtered ($\tau = t$), predicted ($\tau = t-1$), smoothed ($\tau = T$)
$\pi_k(t) = \mathbb{P}(z_{t+1}=k \mid \mathcal{F}_t)$	the gate: weight of expert k for the day- $(t+1)$ return
$\mu_k(x) = f_0(x) + c_k(x; \psi_k)$	mean of expert k : frozen base plus correction
s_k	noise standard deviation of expert k
γ	responsibility: posterior probability of a regime or expert after seeing the returns

Notation clashes to keep in mind

- Bishop writes the *target* as t ; here t is always time.
- `Model_Derivation_BaseCase.md` uses s_t for the regime, $\mathbf{P}$ for the transition matrix, r_t for the market series, μ_k for the gate mean, and r_k for the expert corrections. In these notes those are z_t, A, m_t, ν_k and c_k , which frees r for returns and μ for expert means.
- σ_k is always the *gate's* regime volatility of the market series. s_k is always the *expert's* noise. They measure different things (Section 3.3).
- Chamroukhi & Nguyen use π both for the HMM initial distribution and for the logistic gate. They use α_k for constant mixing proportions, τ for cluster posteriors, γ for regime posteriors, and ξ for pairwise posteriors.

1 From Bishop’s mixture of experts to our model

1.1 Bishop’s starting point and the hidden latent variable

Bishop (§14.5.3) writes a mixture of experts as

$$p(y \mid x) = \sum_{k=1}^K \pi_k(x) p_k(y \mid x). \quad (1)$$

(Bishop writes the target as t ; we write it as y .) No latent variable appears in (1), because it has already been *summed out*. Introduce a one-hot vector z whose component $z_k = 1$ means “expert k generated this observation”. The model then consists of two statements:

$$p(z_k = 1 \mid x) = \pi_k(x), \quad p(y \mid x, z_k = 1) = p_k(y \mid x). \quad (2)$$

Using the exponent trick of Bishop (9.10)–(9.11), the joint distribution is

$$p(y, z \mid x) = p(z \mid x) p(y \mid x, z) = \prod_{k=1}^K [\pi_k(x) p_k(y \mid x)]^{z_k}. \quad (3)$$

Because z is never observed, we sum it out. Each value of z switches on exactly one factor, so

$$p(y \mid x) = \sum_z p(y, z \mid x) = \sum_{k=1}^K \pi_k(x) p_k(y \mid x). \quad (4)$$

The latent variable is therefore still present in (1): π_k is its prior and p_k is the distribution of y given its value. It becomes explicit again in two places:

- in the **complete-data likelihood** that EM works with;
- in the **posterior** (the responsibility)

$$\gamma_k = p(z_k = 1 \mid y, x) = \frac{\pi_k(x) p_k(y \mid x)}{\sum_j \pi_j(x) p_j(y \mid x)}. \quad (5)$$

1.2 Three changes take Bishop’s model to ours

	Bishop	Ours	Meaning
Index	n	(i, t)	one observation: stock i on day t
Target	y_n	$r_{i,t+1}$	forward return
Expert input	x_n	$x_{i,t}$	characteristics of the stock
Gate input	x_n	$\mathcal{F}_t$	market information up to t
Latent	z_n , one per point	z_{t+1} , one per date	the regime, shared by all stocks
Expert mean	$w_k^\top \phi(x)$	$\mu_k(x_{i,t}; \theta_k)$	linear regression becomes an MLP (Section 3)
Noise	β^{-1} , shared	s_k^2 , per expert	each regime has its own noise level
Gate	$\text{softmax}_k(v^\top x_n)$	HMM probability	Sections 4 and 5

Change 1: the indices. Bishop has a single index n . Our data form a panel, so we use two indices, stock i and day t .

Change 2: the experts. Bishop’s expert is already a regression whose mean depends on the input. The MLP makes that mean function flexible. Section 3 shows that an MLP is linear regression on learned basis functions.

Change 3: the gate. This change has two parts.

- *What the gate looks at.* Our gate sees only market information $\mathcal{F}_t$, so every stock on the same day receives the same weights. The gate answers “what state is the market in?”, not “what kind of stock is this?”.
- *Where the probability comes from.* Our gate is a probability computed from an explicit model of the regime: a Markov chain over regimes, observed through the market series. It therefore has memory.

The result, with every time index written explicitly, is

$$p(r_{i,t+1} \mid x_{i,t}, \mathcal{F}_t) = \sum_{k=1}^K \underbrace{p(z_{t+1} = k \mid \mathcal{F}_t)}_{\text{HMM gate}} \underbrace{\mathcal{N}(r_{i,t+1}; \mu_k(x_{i,t}; \theta_k), s_k^2)}_{\text{expert } k} \quad (6)$$

Everything to the right of the conditioning bar is known at the close of day t . Only $r_{i,t+1}$ lies in the future. For the study’s 5-day target the gate weight is the average of this probability over the window, which reduces to it when $h = 1$ (Section 8.9, Q27).

1.3 The generative story

For each day:

1. the market regime takes one step along its Markov chain;
2. the market return for that day is drawn from that regime’s distribution (this is what the HMM sees);
3. every stock’s return is drawn from the expert of the same regime, with a mean set by the stock’s own characteristics.

We never observe step 1. The gate is our best guess of it given the market history, and the mixture averages the experts over that guess.

Why an HMM gate rather than Bishop’s learned softmax?

- *Persistence.* Regimes last for weeks or months, and the transition matrix builds that memory in.
- *Interpretability.* Each state has an economic meaning (for example calm versus stressed), with its own mean and volatility.
- *Few parameters.* A Gaussian HMM with $K = 2$ has seven parameters (one initial probability, two transition probabilities, two means, two variances), whereas a learned gate is easy to overfit when the signal-to-noise ratio is low.
- *Market information enters only through the gate.* Ye & Borde (2026) found that regime variables helped when they entered through the gate and hurt when they were added to the predictors.

1.4 Information sets as σ -algebras and filtrations

“Market information at time t ” has a precise meaning in probability theory, and that meaning is what makes “no look-ahead” a checkable statement.

The probability space. All random quantities (regimes z_t , market returns m_t , characteristics $x_{i,t}$, stock returns $r_{i,t+1}$) live on one probability space $(\Omega, \mathcal{A}, \mathbb{P})$. Here Ω is the set of possible histories, $\mathcal{A}$ is a σ -algebra of events, and $\mathbb{P}$ is the probability measure.

The gate’s information. The information contained in the market series up to day t is the σ -algebra it generates,

$$\mathcal{F}_t = \sigma(m_1, \dots, m_t) \subseteq \mathcal{A}. \quad (7)$$

This is the smallest σ -algebra with respect to which every m_s , $s \leq t$, is measurable. An event belongs to $\mathcal{F}_t$ exactly when an observer who has seen $m_1, \dots, m_t$ can tell whether it has occurred.

A filtration. Because $\mathcal{F}_s \subseteq \mathcal{F}_t$ whenever $s \leq t$, the family $\mathbb{F} = (\mathcal{F}_t)_{t \geq 1}$ is a *filtration*: information accumulates and is never forgotten. It is the *natural filtration* of the market series. The full information set at the close of t is a larger filtration,

$$\mathcal{G}_t = \sigma(m_s, x_{i,s}, r_{i,s} : s \leq t, i \in \mathcal{S}_s), \quad \mathcal{F}_t \subseteq \mathcal{G}_t. \quad (8)$$

The base-case gate is *designed* to use only $\mathcal{F}_t$. The experts’ inputs $x_{i,t}$ are $\mathcal{G}_t$ -measurable.

The gate as a conditional probability given a σ -algebra.

$$\pi_k(t) = \mathbb{P}(z_{t+1} = k \mid \mathcal{F}_t) = \mathbb{E}[\mathbf{1}\{z_{t+1} = k\} \mid \mathcal{F}_t]. \quad (9)$$

This is a random variable, and it is $\mathcal{F}_t$ -measurable. By the Doob–Dynkin lemma it is therefore a measurable function of the observed path: $\pi_k(t) = g_k(m_1, \dots, m_t)$. The forward recursion of Section 5 is the explicit formula for g_k .

No look-ahead means adaptedness. A sequence of forecasts $(\hat{r}_{i,t+1})_t$ is free of look-ahead exactly when it is *adapted* to the information filtration, i.e. $\hat{r}_{i,t+1}$ is $\mathcal{G}_t$ -measurable for every t . The gate must be $\mathcal{F}_t$ -measurable. The smoothed probability $\xi_{t|T}(k) = \mathbb{P}(z_t = k \mid \mathcal{F}_T)$ is $\mathcal{F}_T$ -measurable but in general *not* $\mathcal{F}_t$ -measurable, because it depends on $m_{t+1}, \dots, m_T$. That is the formal version of “smoothed probabilities leak the future” (Section 8).

Notation. Wherever these notes write $p(\cdot \mid \mathcal{F}_t)$ or $p(\cdot \mid m_{1:t})$, they mean the same object: conditioning on $\mathcal{F}_t$ is conditioning on the values of $m_1, \dots, m_t$.

1.5 Two notational points

The bar “|” versus the semicolon “;”.

- $p(a \mid b)$ conditions on a *random variable*: b has a distribution, and $p(a \mid b) = p(a, b)/p(b)$.
- $p(a; \theta)$ indexes a family of distributions by a *fixed parameter*. Since $p(\theta)$ is not defined, Bayes’ rule cannot be applied to θ .
- Frequentist papers such as Jordan & Jacobs (1994) write $\mathcal{N}(y; \mu, \sigma^2)$. Bishop writes | for both cases.

A careful version of our expert density puts each quantity where it belongs:

$$p(r_{i,t+1} \mid x_{i,t}, z_{t+1} = k; \theta_k, s_k) = \mathcal{N}(r_{i,t+1}; \mu_k(x_{i,t}; \theta_k), s_k^2). \quad (10)$$

The rule: anything that will ever be summed over or appear on the left of a probability is a random variable and goes after $\mid$.

The target is a future return, not the next feature vector.

- $x_{i,t+1}$ is tomorrow’s *characteristic vector*, whereas the target is a single number, the return $r_{i,t+1}$. In Weigend et al. (1995) a series predicts its own next value, which is why they write the target as the next input. In a cross-sectional model the inputs and the target are different variables.
- Two dating conventions are in use, and both are fine if applied consistently:
 - label by *realisation* date: $y_{i,t} = r_{i,t}$, conditioned on information at $t - 1$;
 - label by *formation* date: $y_{i,t} := r_{i,t+1}$, conditioned on information at t . The code and `Model_Derivation_BaseCase.md` use this one.
- These notes always write the forward return explicitly: $r_{i,t+1}$ is predicted from $x_{i,t}$, the regime that generates it is z_{t+1} , and the market return over the same day is m_{t+1} . That way the timing is visible in every equation.

2 Warm-up: linear-regression experts

With linear experts, EM has closed-form steps, so every part of the machinery (complete-data likelihood, E-step, M-step) can be derived by hand. Sections 3–5 then swap the linear experts for MLPs and the softmax gate for an HMM. The structure of the derivation stays the same; only the M-step loses its closed form.

2.1 Reading guide (about 2.5 hours)

The main reference is F. Chamroukhi and H. D. Nguyen, *Model-Based Clustering and Classification of Functional Data*, arXiv:1803.00276 (2018). Page numbers below are the printed ones.

The paper has *two* layers of latent variables:

- an outer cluster label Z_i for a whole curve i ;
- an inner regime H_{ij} at each point j of the curve.

Keep this in mind while reading.

R1. §2.1–2.2, pp. 8–11 (20 min). The general mixture, the complete-data log-likelihood (eq. 4), and EM in general form (eqs. 5–8). This is the template that every later section fills in.

R2. §3.1–3.2, pp. 14–16 (30 min). A mixture of linear regressions and its EM. The M-step is weighted least squares (eqs. 18–19).

- Each curve i has *one* label Z_i shared by all of its m_i points, so the component density is a product over those points.
- This is exactly our “one regime per date, shared by every stock”: read *curve* as *date* and *point* as *stock*.

R3. §4.3.1–4.3.2, pp. 48–51 (45 min). The RHLP model, which is a mixture of experts with linear (polynomial) experts and a softmax gate. Read it with one cluster, i.e. ignore the outer mixture. Things to notice:

- the E-step (eq. 54) needs no forward–backward pass;
- the experts are updated by weighted least squares;
- the gate is updated by a weighted multinomial logistic regression, solved by IRLS (iteratively reweighted least squares, which is Newton’s method for logistic regression).

This section is almost exactly Exercise A1.

R4. Bishop, as a companion (45 min).

- §14.5.1: mixtures of linear regression models and their EM.
- §§4.3.3–4.3.4: IRLS and multiclass logistic regression, needed for A1(g).

R5. For Exercise A2.

- §4.2.1–4.2.2, pp. 41–45: the mixture of HMM regressions. The E-step uses forward–backward, and eqs. 39–47 give weighted updates for the initial distribution, the transition matrix, β and σ^2 .
- Bishop §13.2.1–13.2.2 and §13.2.4: the forward–backward algorithm and its scaling factors.

Skip: §3.3–3.6 (regularised mixtures and experiments), §4.1 (piecewise regression) and §5 (discriminant analysis).

Chamroukhi & Nguyen	These notes
α_k (constant mixing proportions)	π_k with no gate input
$\pi_{kr}(x_j; w_k)$ (logistic gate)	$\pi_k(x)$
π_{kr} in §4.2	HMM initial distribution ρ_k
$A_{k\ell r}$	transition matrix A_{jk}
$\tau_{ik}, \gamma_{ijr}, \xi_{ij\ell r}$	responsibilities, regime posteriors, pairwise posteriors
β_k, σ_k^2	expert weights β_k and expert noise s_k^2

2.2 Exercise A1: softmax-gated MoE with linear-regression experts

Exercise A1

Setup. The data are $\mathcal{D} = \{(x_n, y_n)\}_{n=1}^N$ with $x_n \in \mathbb{R}^d$ (first entry 1, the intercept) and $y_n \in \mathbb{R}$. There are K experts and a one-hot latent z_n .

$$p(z_{nk} = 1 \mid x_n; w) = \pi_k(x_n; w) = \frac{\exp(w_k^\top x_n)}{\sum_{j=1}^K \exp(w_j^\top x_n)}, \quad w_K = 0,$$

$$p(y_n \mid x_n, z_{nk} = 1; \beta_k, s_k^2) = \mathcal{N}(y_n; \beta_k^\top x_n, s_k^2),$$

$$\Psi = (w_1, \dots, w_{K-1}, \beta_1, \dots, \beta_K, s_1^2, \dots, s_K^2).$$

- Write the generative story (ancestral sampling for one data point). Draw the graphical model and mark what is observed, what is latent, and where the parameters enter.
- Write the complete-data likelihood $p(Y, Z \mid X; \Psi)$ using 1-of- K exponents, then its

logarithm.

- (c) Write the observed-data log-likelihood $\log p(Y | X; \Psi)$, showing the step where Z is summed out. Why is there no closed-form maximiser?
- (d) E-step: derive $\tau_{nk}^{(q)} = p(z_{nk} = 1 | y_n, x_n; \Psi^{(q)})$ using Bayes' rule.
- (e) Write $Q(\Psi; \Psi^{(q)})$. Show that it splits into one gate term that depends only on w , plus K expert terms, each depending only on (β_k, s_k^2) .
- (f) Expert M-step: derive $\beta_k^{(q+1)}$ and $s_k^{2(q+1)}$ in closed form. Write β_k in matrix form using the design matrix X , the target vector y and a diagonal weight matrix.
- (g) Gate M-step: write the gate objective, its gradient with respect to w_k , and the blocks of its Hessian. Show that the objective is concave, explain why there is no closed form, and write one Newton–Raphson (IRLS) step.
- (h) Special case: the gate has no input, so π_k is a constant α_k . Show that the M-step becomes $\alpha_k^{(q+1)} = \frac{1}{N} \sum_n \tau_{nk}^{(q)}$.
- (i) **The panel version.** Days t , stocks $i \in \mathcal{S}_t$, one latent z_t per date shared by every stock on that date, a gate that depends only on a date-level vector u_t , and experts

$$p(r_{i,t+1} | x_{i,t}, z_{tk} = 1) = \mathcal{N}(r_{i,t+1}; \beta_k^\top x_{i,t}, s_k^2).$$

Redo (b), (c), (d) and (f). What changes in τ , and why does it become much sharper as N_t grows? (Hint: Chamroukhi & Nguyen §3.)

- (j) *(Optional, but it is the key to Section 4.)* Show that

$$\begin{aligned} \log L(\Psi) &= Q(\Psi; \Psi^{(q)}) + H(\Psi; \Psi^{(q)}), \\ H(\Psi; \Psi^{(q)}) &= -\mathbb{E}_{Z \sim p(\cdot | Y, X; \Psi^{(q)})} [\log p(Z | Y, X; \Psi)], \end{aligned}$$

and use this to argue that EM never decreases the log-likelihood.

Solution: Appendix A.

2.3 Exercise A2: the same linear MoE with an HMM gate

Exercise A2 makes three changes to A1: the regime is shared by all stocks on a day, it follows a Markov chain, and it also generates the market series that the gate sees. This is the linear version of our model. Section 5 explains the step from softmax to HMM.

Exercise A2

Setup. Days $t = 1, \dots, T$, stocks $i \in \mathcal{S}_t$, market series m_t , and regime $z_t \in \{1, \dots, K\}$.

$$\begin{aligned} p(z_1 = k) &= \rho_k, & p(z_{t+1} = k | z_t = j) &= A_{jk}, \\ p(m_t | z_t = k) &= \mathcal{N}(m_t; \nu_k, \sigma_k^2), \\ p(r_{i,t+1} | x_{i,t}, z_{t+1} = k) &= \mathcal{N}(r_{i,t+1}; \beta_k^\top x_{i,t}, s_k^2), \end{aligned} \quad \text{independently over } i \in \mathcal{S}_t.$$

Tomorrow's return $r_{i,t+1}$ is predicted from today's characteristics $x_{i,t}$ and generated by tomorrow's regime z_{t+1} , which also generates tomorrow's market return m_{t+1} .

- (a) Draw the graphical model (a chain over z_t ; a plate over stocks). List what is observed, what is latent, what are parameters, and what is only conditioned on.

- (b) Starting from the product rule and the chain rule, derive the joint $p(z_{1:T}, m_{1:T}, R \mid X)$, where R collects all $r_{i,t+1}$. State every conditional-independence assumption you use.
- (c) Define the day- $(t+1)$ emission $e_{t+1}(k) = p(m_{t+1}, \{r_{i,t+1}\}_{i \in \mathcal{S}_t} \mid z_{t+1} = k, X)$ (for $t = 0$ take $e_1(k) = p(m_1 \mid z_1 = k)$). Write the observed-data likelihood as a sum over regime paths. Write the forward recursion (scaled, as in Bishop §13.2.4) and show that it computes the likelihood in $O(TK^2)$ operations.
- (d) Write the backward recursion and the posteriors $\gamma_t(k) = p(z_t = k \mid \text{all data})$ and $\xi_t(j, k) = p(z_t = j, z_{t+1} = k \mid \text{all data})$.
- (e) Write Q and derive the M-step for ρ , A , ν_k , σ_k^2 , β_k and s_k^2 . Compare β_k with Chamroukhi & Nguyen eq. (46).
- (f) **The two-stage version.** First fit (ρ, A, ν, σ) by Baum–Welch on $m_{1:T}$ alone and freeze them. Then fit (β, s) using the market-only predicted probabilities $\pi_k(t) = \xi_{t+1|t}(k)$ as a fixed prior for the day- $(t+1)$ cross-section. Write the Stage-2 E-step and M-step. Which information does the joint fit use that the two-stage fit ignores?
- (g) At the close of day t you want to forecast $r_{i,t+1}$. Which regime probability may you use? Show that using the smoothed posterior γ_{t+1} would be look-ahead.
- (h) Map the model onto Chamroukhi & Nguyen’s mixture of HMM regressions (§4.2): what corresponds to their n , K , R_k , m_i , y_{ij} and W_{ikr} ?

Solution: Appendix B.

3 The experts: from $w^\top \phi(x)$ to an MLP

3.1 What $w^\top \phi(x)$ stands for: linear basis-function regression

Bishop’s experts (and the experts in Exercise A1) are *linear basis-function models* (Bishop §3.1):

$$y = w^\top \phi(x) + \varepsilon = \sum_{j=0}^{M-1} w_j \phi_j(x) + \varepsilon, \quad \phi_0(x) \equiv 1, \quad \varepsilon \sim \mathcal{N}(0, s^2). \quad (11)$$

- $x \in \mathbb{R}^d$ is the raw input.
- $\phi(x) = (\phi_0(x), \dots, \phi_{M-1}(x))^\top$ is a vector of M *fixed* functions of x , the *basis functions* or *features*.
- $w \in \mathbb{R}^M$ is the vector of weights, one per basis function. w_0 is the intercept, because $\phi_0 \equiv 1$.

The model is called *linear* because it is linear in the *weights* w . It need not be linear in x . Common choices of ϕ :

Basis	$\phi_j(x)$
identity (plain linear regression)	$\phi_j(x) = x_j$, so $w^\top \phi(x) = w_0 + \sum_{j=1}^d w_j x_j$
polynomial (one input)	$\phi_j(x) = x^j$
Gaussian radial basis	$\phi_j(x) = \exp(-\ x - c_j\ ^2 / (2\ell^2))$, fixed centres c_j
sigmoidal	$\phi_j(x) = \sigma((x - c_j)/\ell)$, fixed c_j, ℓ

Why it is convenient. Stack the data as the $N \times M$ *design matrix* Φ , whose row n is $\phi(x_n)^\top$. The log-likelihood under (11),

$$\log p(y \mid X; w, s^2) = -\frac{N}{2} \log(2\pi s^2) - \frac{1}{2s^2} \sum_{n=1}^N (y_n - w^\top \phi(x_n))^2, \quad (12)$$

is a *concave quadratic* in w . Setting its gradient to zero gives the normal equations and a closed form:

$$\Phi^\top \Phi \hat{w} = \Phi^\top y \implies \hat{w} = (\Phi^\top \Phi)^{-1} \Phi^\top y. \quad (13)$$

With weights τ_n on the observations, the same derivation gives *weighted* least squares, $\hat{w} = (\Phi^\top W \Phi)^{-1} \Phi^\top W y$ with $W = \text{diag}(\tau_1, \dots, \tau_N)$. This is the expert M-step of Exercise A1.

Why it is not enough. The basis is chosen *before* seeing the data. To approximate an arbitrary smooth function of d inputs with fixed local bases, the number of basis functions must grow exponentially in d (Bishop §1.4, the curse of dimensionality). With dozens of firm characteristics, interactions and thresholds, we cannot guess the right ϕ . The MLP solves this by *learning* ϕ .

3.2 The MLP: linear regression on learned basis functions

One hidden layer. Make each basis function a parametrised “ridge” function of x and learn its parameters:

$$\phi_j(x; W, b) = g\left(\sum_{l=1}^d W_{jl} x_l + b_j\right), \quad j = 1, \dots, H. \quad (14)$$

- g is a fixed scalar nonlinearity, the *activation*.
- W_{jl} and b_j are learned.
- H is the number of hidden units.

The output is linear regression on these learned features:

$$f(x) = b^{(2)} + \sum_{j=1}^H w_j^{(2)} g\left(b_j^{(1)} + \sum_{l=1}^d W_{jl}^{(1)} x_l\right) = w^{(2)\top} \phi(x; W^{(1)}, b^{(1)}) + b^{(2)}. \quad (15)$$

Compare with (11): the last layer of an MLP is exactly $w^\top \phi(x)$. The hidden layers *are* ϕ , and they are fitted to the data instead of being chosen in advance.

L hidden layers, in general form. Write $h^{(0)} = x$ and $H_0 = d$. For $\ell = 1, \dots, L$,

$$a^{(\ell)} = W^{(\ell)} h^{(\ell-1)} + b^{(\ell)}, \quad W^{(\ell)} \in \mathbb{R}^{H_\ell \times H_{\ell-1}}, \quad b^{(\ell)} \in \mathbb{R}^{H_\ell}, \quad (16)$$

$$h^{(\ell)} = g(a^{(\ell)}) \quad (\text{applied elementwise}), \quad (17)$$

$$f(x) = w^{(L+1)\top} h^{(L)} + b^{(L+1)}. \quad (18)$$

- $a^{(\ell)}$ is the *pre-activation* of layer ℓ and $h^{(\ell)}$ its *activation*.
- $h^{(L)}$ is the learned basis vector $\phi(x; W^{(1:L)}, b^{(1:L)})$.

Fully expanded for two hidden layers.

$$f(x) = b^{(3)} + \sum_{j=1}^{H_2} w_j^{(3)} g\left(b_j^{(2)} + \sum_{m=1}^{H_1} W_{jm}^{(2)} g\left(b_m^{(1)} + \sum_{l=1}^d W_{ml}^{(1)} x_l\right)\right). \quad (19)$$

Gu–Kelly–Xiu NN3. NN3 has three hidden layers of widths 32, 16, 8 and ReLU activations:

$$f(x) = b^{(4)} + \sum_{a=1}^8 w_a^{(4)} g\left(b_a^{(3)} + \sum_{b=1}^{16} W_{ab}^{(3)} g\left(b_b^{(2)} + \sum_{c=1}^{32} W_{bc}^{(2)} g\left(b_c^{(1)} + \sum_{l=1}^d W_{cl}^{(1)} x_l\right)\right)\right). \quad (20)$$

Activations.

Name	$g(a)$	$g'(a)$
tanh	$\tanh a$	$1 - \tanh^2 a$
ReLU	$\max(0, a)$	$\mathbf{1}\{a > 0\}$
GELU	$a \Phi_{\mathcal{N}}(a)$ ($\Phi_{\mathcal{N}}$ the standard normal cdf)	$\Phi_{\mathcal{N}}(a) + a \varphi_{\mathcal{N}}(a)$

Why the nonlinearity is essential. If $g(a) = a$, composing the layers collapses to a single linear map:

$$W^{(3)}(W^{(2)}(W^{(1)}x + b^{(1)}) + b^{(2)}) + b^{(3)} = \tilde{W}x + \tilde{b}, \quad \tilde{W} = W^{(3)}W^{(2)}W^{(1)}. \quad (21)$$

Depth then buys nothing. With a ReLU, each hidden unit is a hinge, flat on one side of the hyperplane $W_j \cdot x + b_j = 0$ and linear on the other. A weighted sum of hinges is a piecewise-linear function whose kinks the network places where the data need them. More units give finer pieces, and deeper layers recombine the pieces. This is the intuition behind the universal approximation theorems (Cybenko 1989; Hornik, Stinchcombe & White 1989): a single hidden layer with enough units can approximate any continuous function on a compact set arbitrarily well.

Parameter count.

$$\# \text{params} = \sum_{\ell=1}^{L+1} (H_{\ell-1} + 1) H_{\ell}, \quad H_{L+1} = 1. \quad (22)$$

Example: NN3 with $d = 20$ inputs has $21 \cdot 32 + 33 \cdot 16 + 17 \cdot 8 + 9 \cdot 1 = 672 + 528 + 136 + 9 = 1345$ parameters.

What is lost compared with linear experts. The log-likelihood is no longer concave in the parameters, because $W^{(1:L)}$ sit inside g . Three consequences:

1. There is no closed form, so the parameters must be fitted iteratively by gradient methods (Section 4).
2. There are many local optima, so the result depends on the initial values and on the random seed.
3. The parameters are not identifiable: permuting the hidden units of a layer leaves f unchanged, and for $\tanh$ so does flipping the signs of a unit's incoming and outgoing weights. Each layer therefore has $H_{\ell}! 2^{H_{\ell}}$ equivalent weight settings (Bishop §5.1.1). This is harmless for prediction, but it means individual weights cannot be interpreted.

Our experts. Expert k has mean

$$\mu_k(x; \theta_k) = f_0(x; \phi) + c_k(x; \psi_k), \quad (23)$$

- f_0 is the frozen base MLP;
- c_k is an MLP correction whose final layer is initialised to zero, so the model starts exactly at the base;
- $\theta_k = (\phi, \psi_k)$, with ϕ frozen.

3.3 Why s_k is the noise: what the MLP models and what it does not

Write expert k as a regression with additive Gaussian noise:

$$r_{i,t+1} = \mu_k(x_{i,t}) + s_k \varepsilon_{i,t+1}, \quad \varepsilon_{i,t+1} \sim \mathcal{N}(0, 1) \text{ i.i.d.} \quad (24)$$

Equivalently $r_{i,t+1} \mid x_{i,t}, z_{t+1} = k \sim \mathcal{N}(\mu_k(x_{i,t}), s_k^2)$, so that

$$\mu_k(x) = \mathbb{E}[r \mid x, z = k], \quad s_k^2 = \text{Var}(r \mid x, z = k) = \mathbb{E}[(r - \mu_k(x))^2 \mid x, z = k]. \quad (25)$$

- **The MLP models only the conditional mean:** the part of tomorrow's return that the characteristics predict in regime k .
- **s_k models everything the characteristics cannot explain:** firm news, liquidity shocks, the unpredictable part of returns. It is the scatter of realised returns around the expert's prediction.
- **s_k does not depend on x .** The model assumes the noise has the same size for every stock in regime k (homoskedasticity within a regime). The MLP produces one number, the mean, so the scale must be a separate parameter. A version in which the network also outputs $s_k(x)$ exists (Nix & Weigend 1995) but is not used here.

Why it behaves like a volatility. In a calm regime, returns cluster tightly around what characteristics predict, so s_k is small. In a stressed regime, cross-sectional dispersion rises and the residuals become large, so s_k is large. In finance language, s_k is the regime- k *residual (idiosyncratic) volatility of the cross-section around the model*.

It is *not* the gate's σ_k , which is the time-series volatility of the market return in regime k . The two will tend to move together, but they measure different things.

What s_k does in the likelihood. The log-density of one observation is

$$\log \mathcal{N}(r; \mu_k, s_k^2) = -\frac{1}{2} \log 2\pi - \log s_k - \frac{(r - \mu_k)^2}{2s_k^2}. \quad (26)$$

1. *With s_k fixed*, maximising (26) over μ_k is exactly least squares: s_k only rescales the squared error. This is why “Gaussian NLL = MSE when σ is fixed”.
2. *With μ_k fixed*, setting the derivative with respect to s_k^2 to zero shows that s_k^2 equals the mean squared residual. In a mixture this becomes a responsibility-weighted mean: Exercise A1(f), and Weigend et al. eq. 22.
3. *In a mixture, s_k decides who owns a data point*. A large residual is far more plausible under a wide expert than under a narrow one, so a crash-day return is claimed by the high- s_k expert even if both means are similar. Weigend et al. (footnote 12) report that with all variances clamped equal, accuracy was similar but the regimes were no longer discovered reliably.
4. *The term $-\log s_k$ penalises width*. Without it, the likelihood could be improved simply by making every s_k huge. The opposite failure also exists: an expert that wins only a few points can drive $s_k \rightarrow 0$ and the likelihood to $+\infty$. Weigend et al. guard against this with a prior that shrinks s_k^2 towards a reference value (eq. 23). The code currently has only a 100-step freeze of s_k (`sigma_freeze_steps`).

4 The softmax gate, its likelihood, EM and backpropagation

This section treats the textbook gate (Bishop’s softmax) together with MLP experts. Section 5 then turns the gate into an HMM. The machinery built here (complete-data likelihood, EM, the gradient identity, backpropagation) carries over unchanged.

4.1 Where the softmax gate comes from

Derivation 1: Bayes’ rule with Gaussian class-conditional densities (Bishop §4.2.2).

Suppose the gate input u is generated by the regime: first draw k with probability $p(k)$, then $u \sim \mathcal{N}(u; m_k, \Sigma)$ with a covariance Σ shared by all regimes. Bayes’ rule gives

$$p(k | u) = \frac{p(u | k) p(k)}{\sum_j p(u | j) p(j)} = \frac{\exp(a_k)}{\sum_j \exp(a_j)}, \quad a_k = \log [p(u | k) p(k)]. \quad (27)$$

Any posterior can be written as a softmax of log-joint probabilities, so this step is only a rewriting. The assumption enters when we expand the Gaussian:

$$a_k = -\frac{1}{2} u^\top \Sigma^{-1} u + m_k^\top \Sigma^{-1} u - \frac{1}{2} m_k^\top \Sigma^{-1} m_k + \log p(k) + \text{const}. \quad (28)$$

The quadratic term $-\frac{1}{2} u^\top \Sigma^{-1} u$ and the constant are the same for every k , so they cancel between numerator and denominator. What remains is linear in u :

$$p(k | u) = \text{softmax}_k (w_k^\top u + w_{k0}), \quad w_k = \Sigma^{-1} m_k, \quad w_{k0} = -\frac{1}{2} m_k^\top \Sigma^{-1} m_k + \log p(k). \quad (29)$$

A softmax of linear functions is therefore the exact posterior of a generative model of the gate input. If the covariances differ across regimes, the quadratic terms no longer cancel and the logits become quadratic in u . More generally, any exponential-family class-conditional density with a shared dispersion parameter gives linear logits (Bishop §4.2.4).

Derivation 2: linear log-odds (the multinomial logit). Assume directly that the log-odds of regime k against a reference regime K are linear in the input:

$$\log \frac{p(k | u)}{p(K | u)} = w_k^\top u \quad (k = 1, \dots, K-1). \quad (30)$$

Exponentiate, impose $\sum_k p(k | u) = 1$, and set $w_K = 0$:

$$p(k | u) = \frac{\exp(w_k^\top u)}{1 + \sum_{j < K} \exp(w_j^\top u)} = \frac{\exp(w_k^\top u)}{\sum_{j=1}^K \exp(w_j^\top u)}. \quad (31)$$

This is the canonical-link generalised linear model for a categorical outcome.

Identifiability. Adding the same vector c to every w_k multiplies the numerator and the denominator by $\exp(c^\top u)$, so π is unchanged. Only $K-1$ of the w_k can be identified, which is why one of them is fixed at zero.

Why it is called “soft max”. With a temperature τ , $\text{softmax}_k(a/\tau)$ tends to the one-hot vector of $\arg \max_k a_k$ as $\tau \rightarrow 0$ and to the uniform distribution as $\tau \rightarrow \infty$. Its derivative, which we use repeatedly below, is

$$\frac{\partial \pi_j}{\partial a_k} = \pi_j (\delta_{jk} - \pi_k). \quad (32)$$

The bridge to Section 5

The HMM filter of Section 5 is also a softmax. Its logits are the log of the emission density plus the log of the prior that the transition matrix carries forward from yesterday:

$$\xi_{t|t}(k) = \text{softmax}_k \left(\log \eta_t(k) + \log \sum_j A_{jk} \xi_{t-1|t-1}(j) \right).$$

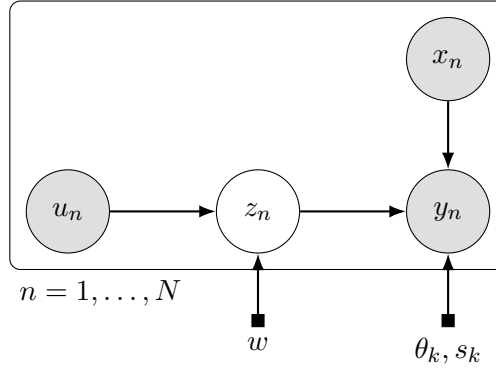
In the softmax gate, the “prior” part of the logit is a fixed bias w_{k0} . In the HMM it is recomputed every day from yesterday’s belief. That is the precise sense in which the HMM gate is a softmax gate with memory.

4.2 The generative model of the softmax-gated MLP MoE

The data are $\{(u_n, x_n, y_n)\}_{n=1}^N$: u_n is the gate input (in Bishop $u_n = x_n$), x_n the expert input and y_n the target.

$$z_n \mid u_n \sim \text{Cat}(\pi_1(u_n; w), \dots, \pi_K(u_n; w)), \quad \pi_k(u; w) = \text{softmax}_k(w_k^\top u), \quad (33)$$

$$y_n \mid x_n, z_{nk} = 1 \sim \mathcal{N}(\mu_k(x_n; \theta_k), s_k^2), \quad \mu_k \text{ an MLP (Section 3)}. \quad (34)$$



Complete-data likelihood. With one-hot z_n ,

$$p(Y, Z \mid X, U; \Psi) = \prod_{n=1}^N \prod_{k=1}^K \left[\pi_k(u_n; w) \mathcal{N}(y_n; \mu_k(x_n; \theta_k), s_k^2) \right]^{z_{nk}}. \quad (35)$$

Observed-data log-likelihood. Summing out each z_n (Appendix A(c) shows the step) gives

$$\ell(\Psi) = \log p(Y \mid X, U; \Psi) = \sum_{n=1}^N \log \sum_{k=1}^K \pi_k(u_n; w) \mathcal{N}(y_n; \mu_k(x_n; \theta_k), s_k^2). \quad (36)$$

The sum inside the logarithm couples all the parameters, so there is no closed-form maximiser even when the experts are linear.

4.3 Maximising it with EM

E-step. Using the current parameters $\Psi^{(q)}$, compute the responsibilities

$$\gamma_{nk} = p(z_{nk} = 1 \mid y_n, x_n, u_n; \Psi^{(q)}) = \frac{\pi_k(u_n) \mathcal{N}(y_n; \mu_k(x_n), s_k^2)}{\sum_j \pi_j(u_n) \mathcal{N}(y_n; \mu_j(x_n), s_j^2)}. \quad (37)$$

The Q -function is the complete-data log-likelihood with each z_{nk} replaced by γ_{nk} . It separates into a gate part and one part per expert:

$$Q(\Psi; \Psi^{(q)}) = \underbrace{\sum_n \sum_k \gamma_{nk} \log \pi_k(u_n; w)}_{Q_{\text{gate}}(w)} + \underbrace{\sum_k \sum_n \gamma_{nk} \left[-\frac{1}{2} \log(2\pi s_k^2) - \frac{(y_n - \mu_k(x_n; \theta_k))^2}{2s_k^2} \right]}_{Q_k(\theta_k, s_k)}. \quad (38)$$

M-step.

- *Gate.* Q_{gate} is a cross-entropy with soft targets γ_{nk} , i.e. a weighted multinomial logistic regression. It is concave in w and is solved by IRLS (Appendix A(g)).
- *Experts.* For fixed s_k , maximising Q_k over θ_k is *weighted least squares*, $\min_{\theta_k} \sum_n \gamma_{nk} (y_n - \mu_k(x_n; \theta_k))^2$. For linear experts this has a closed form. For MLPs it does not, so one takes a few gradient steps. Doing so still increases Q , which is all EM needs: this is *generalised EM*.
- *Noise.* Setting $\partial Q_k / \partial s_k^2 = 0$ gives $s_k^2 = \sum_n \gamma_{nk} (y_n - \mu_k(x_n))^2 / \sum_n \gamma_{nk}$.

4.4 Why EM never decreases the likelihood

For *any* distribution $q(Z)$ over the latent variables,

$$\begin{aligned} \log p(Y | \Psi) &= \sum_Z q(Z) \log p(Y | \Psi) = \sum_Z q(Z) \log \frac{p(Y, Z | \Psi)}{p(Z | Y, \Psi)} \\ &= \underbrace{\sum_Z q(Z) \log \frac{p(Y, Z | \Psi)}{q(Z)}}_{\mathcal{L}(q, \Psi)} + \underbrace{\sum_Z q(Z) \log \frac{q(Z)}{p(Z | Y, \Psi)}}_{\text{KL}(q \| p(Z | Y, \Psi)) \geq 0}. \end{aligned} \quad (39)$$

So $\mathcal{L}(q, \Psi)$ is a lower bound on the log-likelihood (Bishop §9.4).

- The **E-step** sets $q(Z) = p(Z | Y, \Psi^{(q)})$. The KL term becomes zero and the bound touches the log-likelihood at $\Psi^{(q)}$.
- With this q , $\mathcal{L}(q, \Psi) = Q(\Psi; \Psi^{(q)}) - \sum_Z q \log q$. The second term does not depend on Ψ , so the **M-step**, which maximises Q , maximises the bound.

Therefore

$$\log p(Y | \Psi^{(q+1)}) \geq \mathcal{L}(q, \Psi^{(q+1)}) \geq \mathcal{L}(q, \Psi^{(q)}) = \log p(Y | \Psi^{(q)}). \quad (40)$$

Any $\Psi^{(q+1)}$ that merely *increases* Q keeps this chain of inequalities, which is why generalised EM still works.

4.5 Why gradient training of the likelihood behaves like EM

Fisher's identity. The gradient of the observed-data log-likelihood equals the posterior expectation of the gradient of the complete-data log-likelihood:

$$\nabla_{\Psi} \log p(Y | \Psi) = \frac{\sum_Z \nabla_{\Psi} p(Y, Z | \Psi)}{p(Y | \Psi)} = \sum_Z \frac{p(Y, Z | \Psi)}{p(Y | \Psi)} \nabla_{\Psi} \log p(Y, Z | \Psi) = \mathbb{E}_{p(Z | Y, \Psi)} [\nabla_{\Psi} \log p(Y, Z | \Psi)]. \quad (41)$$

The right-hand side is exactly $\nabla_{\Psi} Q(\Psi; \Psi')$ evaluated at $\Psi' = \Psi$. **A gradient step on the log-likelihood is therefore an E-step (compute γ with the current parameters) followed**

by one small step on the M-step objective. EM and gradient ascent climb the same function and stop at the same kind of point, where the gradient is zero. EM jumps to the maximum of Q when it can; gradient ascent takes small steps. With MLP experts we cannot jump anyway.

The gradients for one observation. Write $\ell_n = \log \sum_k \pi_k \mathcal{N}_k$ with $\mathcal{N}_k = \mathcal{N}(y_n; \mu_k, s_k^2)$.

- *Expert mean.* Only the k -th term depends on μ_k , and $\partial \log \mathcal{N}_k / \partial \mu_k = (y_n - \mu_k) / s_k^2$, so

$$\frac{\partial \ell_n}{\partial \mu_k} = \frac{\pi_k \mathcal{N}_k}{\sum_j \pi_j \mathcal{N}_j} \cdot \frac{y_n - \mu_k}{s_k^2} = \gamma_{nk} \frac{y_n - \mu_k}{s_k^2}. \quad (42)$$

- *Expert noise.* Since $\partial \log \mathcal{N}_k / \partial \log s_k = -1 + (y_n - \mu_k)^2 / s_k^2$,

$$\frac{\partial \ell_n}{\partial \log s_k} = \gamma_{nk} \left[\frac{(y_n - \mu_k)^2}{s_k^2} - 1 \right]. \quad (43)$$

- *Gate logits a_k ,* with $\pi = \text{softmax}(a)$. Using (32),

$$\frac{\partial \ell_n}{\partial a_k} = \sum_j \frac{\mathcal{N}_j}{\sum_l \pi_l \mathcal{N}_l} \pi_j (\delta_{jk} - \pi_k) = \gamma_{nk} - \pi_k \sum_j \gamma_{nj} = \gamma_{nk} - \pi_k. \quad (44)$$

Each gradient is the EM weighting in disguise. The expert that “owns” an observation (large γ_{nk}) receives its error signal, scaled by $1/s_k^2$. The gate is pulled towards the posterior γ , as in Weigend et al. (1995) eqs. 24–25. In our base case the gate is frozen, so only (42) and the noise gradient are used.

4.6 Backpropagation in depth

Backpropagation is the chain rule applied efficiently to the expert MLP of Section 3, eqs. (16)–(18). We minimise the loss $L_n = -\ell_n$ for one observation. Its derivative with respect to expert k ’s output is, from (42),

$$\delta_k^{\text{out}} := \frac{\partial L_n}{\partial \mu_k} = -\gamma_{nk} \frac{y_n - \mu_k}{s_k^2}. \quad (45)$$

Drop the index k below: the same steps run inside every expert.

Step 1: forward pass. Compute and *store* $a^{(\ell)}$ and $h^{(\ell)}$ for $\ell = 1, \dots, L$, then the output μ , the responsibility γ and δ^{out} .

Step 2: output layer. Since $\mu = w^{(L+1)\top} h^{(L)} + b^{(L+1)}$,

$$\frac{\partial L_n}{\partial w^{(L+1)}} = \delta^{\text{out}} h^{(L)}, \quad \frac{\partial L_n}{\partial b^{(L+1)}} = \delta^{\text{out}}. \quad (46)$$

Step 3: define the error at every hidden layer, $\delta^{(\ell)} := \partial L_n / \partial a^{(\ell)} \in \mathbb{R}^{H_\ell}$. For the last hidden layer, μ depends on $a_m^{(L)}$ only through $h_m^{(L)} = g(a_m^{(L)})$, so

$$\delta_m^{(L)} = \delta^{\text{out}} w_m^{(L+1)} g'(a_m^{(L)}), \quad \text{i.e.} \quad \delta^{(L)} = g'(a^{(L)}) \odot w^{(L+1)} \delta^{\text{out}}. \quad (47)$$

Step 4: the backward recursion. Layer ℓ affects the loss only through the pre-activations of layer $\ell + 1$, $a_j^{(\ell+1)} = \sum_m W_{jm}^{(\ell+1)} g(a_m^{(\ell)}) + b_j^{(\ell+1)}$, so $\partial a_j^{(\ell+1)} / \partial a_m^{(\ell)} = W_{jm}^{(\ell+1)} g'(a_m^{(\ell)})$. The chain rule sums over every path from $a_m^{(\ell)}$ to the loss:

$$\delta_m^{(\ell)} = \sum_{j=1}^{H_{\ell+1}} \delta_j^{(\ell+1)} W_{jm}^{(\ell+1)} g'(a_m^{(\ell)}) \iff \delta^{(\ell)} = g'(a^{(\ell)}) \odot \left(W^{(\ell+1)\top} \delta^{(\ell+1)} \right). \quad (48)$$

Step 5: weight gradients. Since $a_j^{(\ell)} = \sum_m W_{jm}^{(\ell)} h_m^{(\ell-1)} + b_j^{(\ell)}$,

$$\frac{\partial L_n}{\partial W_{jm}^{(\ell)}} = \delta_j^{(\ell)} h_m^{(\ell-1)} \iff \frac{\partial L_n}{\partial W^{(\ell)}} = \delta^{(\ell)} h^{(\ell-1)\top}, \quad \frac{\partial L_n}{\partial b^{(\ell)}} = \delta^{(\ell)}. \quad (49)$$

Each gradient is “error arriving at the unit” times “signal that entered the unit”.

A tiny example. One hidden unit, squared-error loss: $f(x) = w_2 g(w_1 x + b_1) + b_2$, $a = w_1 x + b_1$, $L = \frac{1}{2}(y - f)^2$.

$$\frac{\partial L}{\partial w_2} = -(y - f) g(a), \quad \frac{\partial L}{\partial b_2} = -(y - f), \quad \frac{\partial L}{\partial w_1} = -(y - f) w_2 g'(a) x, \quad \frac{\partial L}{\partial b_1} = -(y - f) w_2 g'(a). \quad (50)$$

The error $-(y - f)$ is computed once and reused, travelling backwards through w_2 and g' . That reuse is the whole trick.

Why this is efficient. One backward pass costs about as much as one forward pass and returns the gradient with respect to *all* parameters. Finite differences would need one extra forward pass per parameter. Reverse-mode automatic differentiation (PyTorch’s autograd) carries out exactly Steps 1–5 for any graph we write down, including the mixture log-likelihood and the gate.

The update. Average the gradients over a mini-batch $\mathcal{B}$ and step downhill:

$$\theta \leftarrow \theta - \eta \frac{1}{|\mathcal{B}|} \sum_{n \in \mathcal{B}} \nabla_{\theta} L_n. \quad (51)$$

Adam, which the code uses, rescales each coordinate of this step by running estimates of the gradient’s mean and root mean square. It is still a descent step on the same negative log-likelihood.

Three consequences for our base case.

1. *Frozen base.* f_0 appears inside every μ_k , but its parameters ϕ are excluded from the update, so the error stops at f_0 .
2. *Zero-initialised final layer.* At step zero $w^{(L+1)} = 0$, so $\delta^{(L)} = 0$ and the hidden layers receive no gradient. They start learning one step later, once the final layer has moved.
3. *Mini-batches under the per-date likelihood.* The per-date likelihood (Section 7) couples all stocks of a date through the shared regime, so γ must be computed over *whole dates*. Mini-batches must therefore consist of complete dates. Under the per-row likelihood, which the study uses (Q24), any set of rows will do.

5 From the softmax gate to the HMM gate

5.1 Three steps from Bishop's gate to ours

1. **The gate input becomes date-level.** Instead of the stock's own input, the gate sees the market history $\mathcal{F}_t$.
2. **The latent becomes one per date and is shared.** A single z_{t+1} governs every stock's return on day $t + 1$ (Exercise A1(i), Chamroukhi & Nguyen §3).
3. **The latent gets memory.** The prior of z_{t+1} depends on z_t through a transition matrix A , which makes the gate an HMM.

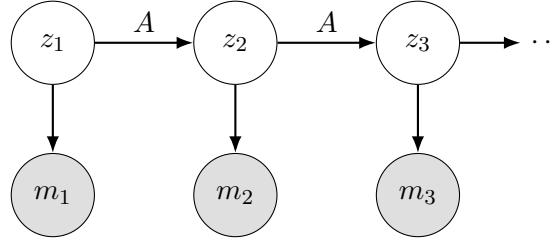
5.2 Why the Markov chain sits on the latent variables (Bishop §13.1–13.2)

Bishop §13.1: a Markov chain on the observations. The simplest way to give a series memory is

$$p(m_{t+1} \mid m_1, \dots, m_t) = p(m_{t+1} \mid m_t). \quad (52)$$

This remembers a single step. A chain of order M needs a number of parameters exponential in M , which is useless for regimes that last months.

Bishop §13.2: put the chain on a hidden variable. Let the hidden regime follow a first-order Markov chain, and let each observation depend only on the current regime:



The observations are then *not* Markov of any finite order. To predict m_{t+1} one needs the belief about z_t , and that belief depends on the whole history $m_1, \dots, m_t$. The model has long memory with only $K(K - 1)$ transition parameters.

The transition matrix. In Bishop's notation (eq. 13.7), with one-hot regimes,

$$A_{jk} = p(z_{t+1} = k \mid z_t = j), \quad 0 \leq A_{jk} \leq 1, \quad \sum_{k=1}^K A_{jk} = 1, \quad p(z_{t+1} \mid z_t, A) = \prod_{j=1}^K \prod_{k=1}^K A_{jk}^{z_{tj} z_{t+1,k}}. \quad (53)$$

Exactly one z_{tj} and one $z_{t+1,k}$ equal 1, so the double product picks out the single entry for the transition that happened. The first regime has its own distribution (Bishop eq. 13.8, where it is called π ; we write ρ because π is our gate):

$$p(z_1 \mid \rho) = \prod_k \rho_k^{z_1 k}. \quad (54)$$

The gate's parameters are $\theta_g = \{\rho, A, \nu_{1:K}, \sigma_{1:K}\}$, the analogue of Bishop's $\theta = \{\pi, A, \phi\}$ in eq. 13.11. For $K = 2$ that is $1 + 2 + 4 = 7$ numbers.

The HMM as a mixture model whose weights move (Bishop §13.2).

Model	Mixing weight for observation $t + 1$	Depends on
Gaussian mixture (Bishop ch. 9)	π_k , a constant	nothing
Mixture of experts (Bishop §14.5)	$\pi_k(u)$, softmax gate	the input
HMM (Bishop §13.2)	A_{jk} , row j of A	the previous regime
Our HMM gate	$\sum_j A_{jk} \xi_{t t}(j)$	the previous regime, averaged over our belief

5.3 What A does in the gate

If we knew today's regime, tomorrow's mixing weights would be row z_t of A . We do not know it, so we apply the sum rule over z_t and then the Markov property:

$$\pi_k(t) = \mathbb{P}(z_{t+1} = k \mid \mathcal{F}_t) = \sum_{j=1}^K p(z_{t+1} = k \mid z_t = j) \mathbb{P}(z_t = j \mid \mathcal{F}_t) = \sum_{j=1}^K A_{jk} \xi_{t|t}(j) \quad (55)$$

Inside the filter, A is used every day in a *predict* step, followed by a Bayes *update* with the new market return. Write $\eta_t(k) = \mathcal{N}(m_t; \nu_k, \sigma_k^2)$. Then

$$\xi_{t+1|t}(k) = \sum_j A_{jk} \xi_{t|t}(j) \quad (\text{predict}), \quad \xi_{t+1|t+1}(k) = \frac{\xi_{t+1|t}(k) \eta_{t+1}(k)}{\sum_l \xi_{t+1|t}(l) \eta_{t+1}(l)} \quad (\text{update}). \quad (56)$$

So A is the model's prior about how regimes move, and the market data correct it every day. The gate (55) is exactly the predict step.

What A tells you.

- *Persistence*: the diagonal A_{kk} is the probability of staying in a regime.
- *Expected duration*: each day the chain leaves regime k with probability $1 - A_{kk}$, so the time spent in k is geometric. First-step analysis gives $\mathbb{E}[\text{duration}] = 1/(1 - A_{kk})$.
- *Long-run share of time in each regime*: the stationary distribution, $\bar{\rho}A = \bar{\rho}$ with $\sum_k \bar{\rho}_k = 1$.
- *Forecasts h days ahead*: $\mathbb{P}(z_{t+h} = k \mid \mathcal{F}_t) = [\xi_{t|t}^\top A^h]_k$. These converge to $\bar{\rho}$; for $K = 2$ the gap shrinks like λ^h , with $\lambda = A_{11} + A_{22} - 1$ the second eigenvalue.

Worked example.

$$A = \begin{pmatrix} 0.98 & 0.02 \\ 0.05 & 0.95 \end{pmatrix}. \quad (57)$$

- Durations: calm lasts $1/0.02 = 50$ days on average, stressed lasts $1/0.05 = 20$ days.
- Stationary distribution: $0.02 \bar{\rho}_1 = 0.05 \bar{\rho}_2$ gives $\bar{\rho} = (5/7, 2/7) \approx (0.71, 0.29)$.
- Second eigenvalue: $\lambda = 0.93$, so a regime forecast loses half of its distance to the long-run mix in $\ln 0.5 / \ln 0.93 \approx 9.6$ days.

5.4 How A is estimated

Step 1: if the regimes were observed. The probability of a regime path depends on A only through the transition counts n_{jk} :

$$p(z_{1:T} \mid A, \rho) = \rho_{z_1} \prod_{t=1}^{T-1} A_{z_t z_{t+1}} = \rho_{z_1} \prod_{j,k} A_{jk}^{n_{jk}}. \quad (58)$$

Maximise the log subject to each row summing to one, with a Lagrange multiplier λ_j per row:

$$\frac{\partial}{\partial A_{jk}} \left[\sum_{j,k} n_{jk} \log A_{jk} + \sum_j \lambda_j \left(1 - \sum_k A_{jk} \right) \right] = \frac{n_{jk}}{A_{jk}} - \lambda_j = 0 \quad \Rightarrow \quad \hat{A}_{jk} = \frac{n_{jk}}{\sum_l n_{jl}}. \quad (59)$$

So the estimate is a frequency: transitions from j to k divided by visits to j .

Step 2: the regimes are hidden, so use EM (Baum–Welch, Bishop §13.2.1). We replace the counts with their posterior expectations, which the forward–backward algorithm provides (next subsection):

$$\hat{A}_{jk} = \frac{\sum_{t=1}^{T-1} \xi_t(j, k)}{\sum_{t=1}^{T-1} \sum_l \xi_t(j, l)}, \quad \xi_t(j, k) = \mathbb{P}(z_t = j, z_{t+1} = k \mid \mathcal{F}_T). \quad (60)$$

5.5 The forward–backward recursion and how α and β are used

The recursion is needed in **Stage 1**, when the gate is fitted to the market series. Stage 2 does not need it, because there the gate prior is fixed for each date (Section 7).

The two quantities (stated here without derivation; Bishop §13.2.2):

$$\alpha_t(k) = p(m_1, \dots, m_t, z_t = k), \quad \alpha_1(k) = \rho_k \eta_1(k), \quad \alpha_{t+1}(k) = \eta_{t+1}(k) \sum_j \alpha_t(j) A_{jk}, \quad (61)$$

$$\beta_t(k) = p(m_{t+1}, \dots, m_T \mid z_t = k), \quad \beta_T(k) = 1, \quad \beta_t(j) = \sum_k A_{jk} \eta_{t+1}(k) \beta_{t+1}(k). \quad (62)$$

α looks at the past and today; β looks at the future.

The five uses.

1. *The likelihood:*

$$p(m_{1:T}) = \sum_k \alpha_T(k) = \sum_k \alpha_t(k) \beta_t(k) \quad \text{for every } t. \quad (63)$$

The second form gives a useful check that the code is correct.

2. *The filtered probability (the gate's ingredient):* $\xi_{t|t}(k) = \alpha_t(k) / \sum_j \alpha_t(j)$.

3. *The gate:* $\pi_k(t) = \sum_j A_{jk} \xi_{t|t}(j)$, eq. (55).

4. *The smoothed probability of a single regime:* $\gamma_t(k) = \mathbb{P}(z_t = k \mid \mathcal{F}_T) = \alpha_t(k) \beta_t(k) / p(m_{1:T})$.

5. *The smoothed probability of a transition:*

$$\xi_t(j, k) = \frac{\overbrace{\alpha_t(j)}^{\text{past, end in } j} \overbrace{A_{jk}}^{\text{jump}} \overbrace{\eta_{t+1}(k)}^{\text{tomorrow in } k} \overbrace{\beta_{t+1}(k)}^{\text{later data from } k}}{p(m_{1:T})}. \quad (64)$$

The Baum–Welch loop.

1. *Initialise* ρ, A, ν, σ , e.g. A with 0.9 on the diagonal and the rest spread evenly.
2. *E-step*: forward and backward passes, then $\gamma_t(k)$ and $\xi_t(j, k)$.
3. *M-step*, all in closed form:

$$\rho_k = \gamma_1(k), \quad A_{jk} = \frac{\sum_{t=1}^{T-1} \xi_t(j, k)}{\sum_{t=1}^{T-1} \gamma_t(j)}, \quad \nu_k = \frac{\sum_t \gamma_t(k) m_t}{\sum_t \gamma_t(k)}, \quad \sigma_k^2 = \frac{\sum_t \gamma_t(k) (m_t - \nu_k)^2}{\sum_t \gamma_t(k)}. \quad (65)$$

4. *Monitor* $\log p(m_{1:T})$, which EM never decreases (Section 4.4). Stop when it stops changing.
5. *Restart* from several initial values and keep the run with the highest likelihood.

Scaling (Bishop §13.2.4). α_t is a product of t densities and underflows. Work with normalised quantities instead:

$$\hat{\alpha}_t(k) = \xi_{t|t}(k), \quad c_t = p(m_t \mid \mathcal{F}_{t-1}), \quad \log p(m_{1:T}) = \sum_{t=1}^T \log c_t, \quad (66)$$

$$\hat{\beta}_T(k) = 1, \quad \hat{\beta}_t(j) = \frac{\sum_k A_{jk} \eta_{t+1}(k) \hat{\beta}_{t+1}(k)}{c_{t+1}}, \quad (67)$$

$$\gamma_t(k) = \hat{\alpha}_t(k) \hat{\beta}_t(k), \quad \xi_t(j, k) = \frac{\hat{\alpha}_t(j) A_{jk} \eta_{t+1}(k) \hat{\beta}_{t+1}(k)}{c_{t+1}}. \quad (68)$$

The normalised forward pass is exactly the Hamilton filter. The Kim smoother used in `Model_Derivation_BaseCase.md` is another way of computing the same γ_t . Packages such as `statsmodels` maximise the same likelihood, with a few EM iterations followed by a numerical optimiser.

α is legal for forecasting, β is not

β_t contains $m_{t+1}, \dots, m_T$, so γ_t and $\xi_t(j, k)$ are $\mathcal{F}_T$ -measurable but not $\mathcal{F}_t$ -measurable (Section 1.4). They may be used only inside the M-step on the *training block*. The walk-forward procedure is therefore: fit Baum–Welch on the training block, freeze θ_g , then run *only the forward pass* through validation and test.

5.6 Initialising and labelling

- *Simulating data*. Choose A yourself (as in the example) and draw z_{t+1} from the categorical distribution given by row z_t .
- *Starting values for EM*. Use a heavy diagonal or random rows, with several restarts.
- *Label switching*. Permuting the state labels permutes the rows and columns of A and leaves the likelihood unchanged. To make “state 1” mean the same thing in every walk-forward fold, order the states after fitting, e.g. by σ_k from low to high. The HMM is fitted first, so this ordering also fixes which expert is which.

6 The full generative model (base case)

6.1 The generative story

Days run $t = 1, \dots, T$. On each day the universe $\mathcal{S}_t$ is fixed at the close, characteristics $x_{i,t}$ are observed, and tomorrow's returns $r_{i,t+1}$ are the targets.

1. **Regime.** $z_1 \sim \text{Cat}(\rho)$. For $t \geq 1$, tomorrow's regime is drawn from row z_t of A :

$$z_{t+1} \mid z_t = j \sim \text{Cat}(A_{j1}, \dots, A_{jK}). \quad (69)$$

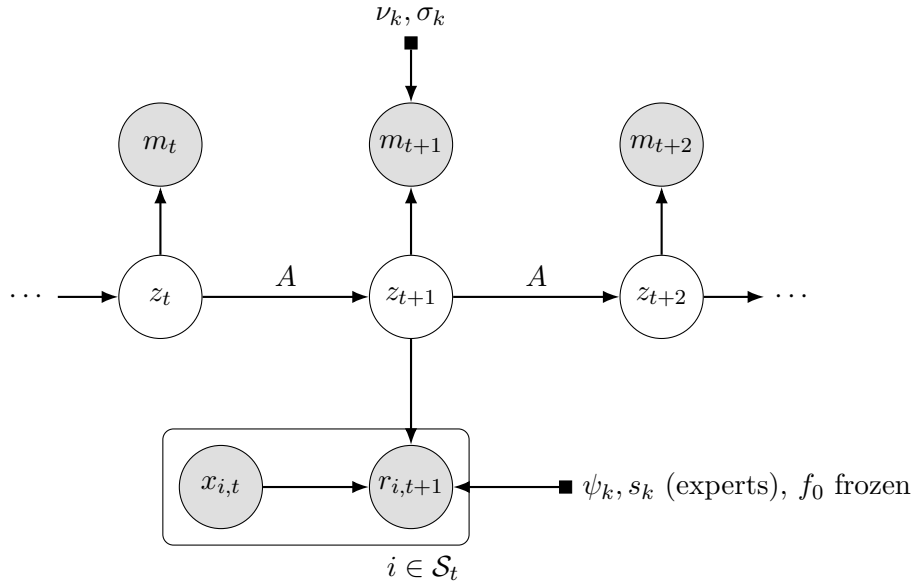
2. **Market.** The market return is drawn from the regime's Gaussian:

$$m_t \mid z_t = k \sim \mathcal{N}(\nu_k, \sigma_k^2). \quad (70)$$

3. **Cross-section.** For every $i \in \mathcal{S}_t$, tomorrow's return is drawn from the expert of tomorrow's regime, with a mean set by the stock's characteristics today:

$$r_{i,t+1} \mid z_{t+1} = k, x_{i,t} \sim \mathcal{N}(\mu_k(x_{i,t}), s_k^2), \quad \mu_k(x) = f_0(x) + c_k(x; \psi_k). \quad (71)$$

6.2 The graph



The model has two layers.

- **The HMM layer:** a chain over days, in which each regime emits that day's market return.
- **The cross-section layer:** a plate over stocks. Each stock's return has two parents, the *shared* regime of its day and its *own* characteristics.

Equivalently, this is an HMM whose emission on day $t + 1$ is the whole vector

$$(m_{t+1}, r_{1,t+1}, \dots, r_{N_t,t+1}).$$

It is Chamroukhi & Nguyen's HMM regression (§4.2) with a vector-valued observation at each time point.

6.3 Deriving the joint distribution

Write R for all returns $r_{i,t+1}$ ($t = 1, \dots, T-1$, $i \in \mathcal{S}_t$) and X for all characteristics.

Step 1: product rule (always true):

$$p(z_{1:T}, m_{1:T}, R \mid X) = p(z_{1:T} \mid X) p(m_{1:T} \mid z_{1:T}, X) p(R \mid m_{1:T}, z_{1:T}, X). \quad (72)$$

Step 2: the regime path. Two assumptions:

- **A0 (exogenous characteristics):** $p(z_{1:T} \mid X) = p(z_{1:T})$.
- **A1 (Markov):** $p(z_{t+1} \mid z_{1:t}) = p(z_{t+1} \mid z_t)$.

Using the chain rule and then A1,

$$p(z_{1:T}) = p(z_1) \prod_{t=1}^{T-1} p(z_{t+1} \mid z_{1:t}) \stackrel{A1}{=} p(z_1) \prod_{t=1}^{T-1} A_{z_t z_{t+1}}. \quad (73)$$

Step 3: the market series.

- **A2 (market emission):** given its own day's regime, m_t is independent of all other regimes, all other market returns and the characteristics.

$$p(m_{1:T} \mid z_{1:T}, X) = \prod_{t=1}^T p(m_t \mid m_{1:t-1}, z_{1:T}, X) \stackrel{A2}{=} \prod_{t=1}^T \mathcal{N}(m_t; \nu_{z_t}, \sigma_{z_t}^2). \quad (74)$$

Step 4: the cross-section.

- **A3 (cross-sectional emission):** given z_{t+1} and its own $x_{i,t}$, $r_{i,t+1}$ is independent of (a) the other stocks' returns, (b) the market return m_{t+1} , (c) all other days' returns and (d) the regimes on other days.

$$p(R \mid m_{1:T}, z_{1:T}, X) \stackrel{A3}{=} \prod_{t=1}^{T-1} \prod_{i \in \mathcal{S}_t} \mathcal{N}(r_{i,t+1}; \mu_{z_{t+1}}(x_{i,t}), s_{z_{t+1}}^2). \quad (75)$$

Result.

$$\boxed{p(z_{1:T}, m_{1:T}, R \mid X) = p(z_1) \prod_{t=1}^{T-1} A_{z_t z_{t+1}} \prod_{t=1}^T \mathcal{N}(m_t; \nu_{z_t}, \sigma_{z_t}^2) \times \prod_{t=1}^{T-1} \prod_{i \in \mathcal{S}_t} \mathcal{N}(r_{i,t+1}; \mu_{z_{t+1}}(x_{i,t}), s_{z_{t+1}}^2)} \quad (76)$$

This is Bishop's eq. 13.10 with one extra product over stocks inside each day. It can be read straight off the graph: one factor per node, each conditioned on its parents (Bishop eq. 8.5).

6.4 What kind of quantity each symbol is

Role	Quantities
Observed	market series $m_{1:T}$, stock returns $r_{i,t+1}$
Conditioned on, not modelled	characteristics $x_{i,t}$, universe $\mathcal{S}_t$
Latent	regimes $z_{1:T}$
Gate parameters θ_g	ρ, A, ν_k, σ_k
Base parameters (frozen)	ϕ inside f_0
Expert parameters	ψ_k inside c_k , noise s_k

The cross-section part of the model is *discriminative*: it models returns given characteristics, never the characteristics themselves. The market part is *generative*: it models the series itself.

6.5 How realistic the assumptions are

- **A0** is mild. Characteristics such as momentum do vary with the regime, but we condition on them rather than use them to infer it. This is the design choice that keeps stock information out of the gate.
- **A1** carries the persistence. It implies geometric regime durations (Chamroukhi & Nguyen §4.3).
- **A2** holds for a plain Gaussian Hamilton model. The autoregressive version deliberately breaks it, which is the origin of the alignment bug G-2.
- **A3(b)** is the weak point for *raw* returns. The market return is roughly the average of the stock returns, so on a -4% day almost every stock is down beyond what the regime explains. With the *market-neutral* target the study uses (Q25) the common shock is removed and A3(b) becomes a reasonable approximation.
- **A3(a)** is only approximate, because industry peers stay correlated. This would make the per-date posterior of Section 7 overconfident, and is one reason the experts are trained on the per-row composite likelihood instead (Section 7.3, Q24).
- **A4 (homoskedastic experts)**: s_k is the same for every stock in regime k (Section 3.3).

Base-case estimation (Section 7). The gate parameters are fitted by maximum likelihood on the market series (Baum–Welch) and frozen. The expert parameters are then fitted by maximising the mixture likelihood of the returns. There are no penalties, no tempering and no joint fit.

7 The likelihood (base case)

7.1 What is maximised, and in what order

The full model's likelihood is (76) with every regime summed out. The base case does not maximise this in one go. It maximises two pieces in sequence:

Stage	Maximises	Parameters	Algorithm	Afterwards
1	$\log p(m_{1:T}; \theta_g)$	ρ, A, ν, σ	Baum–Welch (§5.5)	frozen
2	log-likelihood of the returns, gate fixed	ψ_k, s_k	gradient ascent (§7.3)	—

The Stage 1 objective is computed by the scaled forward pass:

$$\ell_{\text{gate}}(\theta_g) = \log p(m_{1:T}; \theta_g) = \sum_{t=1}^T \log c_t. \quad (77)$$

7.2 The Stage 2 likelihood: summing out the regime date by date

Fix a date t . Tomorrow's cross-section depends on the unknown z_{t+1} . Apply the sum rule over z_{t+1} and then A3, which makes the stocks independent given the regime:

$$\begin{aligned} p(r_{\cdot, t+1} \mid x_{\cdot, t}, \mathcal{F}_t) &= \sum_{k=1}^K \mathbb{P}(z_{t+1} = k \mid \mathcal{F}_t) p(r_{\cdot, t+1} \mid x_{\cdot, t}, z_{t+1} = k) \\ &= \sum_{k=1}^K \pi_k(t) \prod_{i \in \mathcal{S}_t} \mathcal{N}(r_{i, t+1}; \mu_k(x_{i, t}), s_k^2). \end{aligned} \quad (78)$$

Taking logs and adding over dates gives the **per-date likelihood**:

$$\ell_{\text{date}}(\psi, s) = \sum_{t=1}^{T-1} \log \sum_{k=1}^K \pi_k(t) \prod_{i \in \mathcal{S}_t} \mathcal{N}(r_{i, t+1}; \mu_k(x_{i, t}), s_k^2) \quad (79)$$

Why adding over dates is allowed. The gate is frozen and $\mathcal{F}_t$ -measurable, i.e. built from the market alone. So the returns on one date never change the regime belief used on another date, and each date's term stands alone. Formally, (79) is a *cut* (composite) likelihood: the flow of information from returns back into the regime belief is deliberately removed. This is also why Stage 2 needs no forward–backward recursion.

What the full joint fit would do instead (not used). Treat

$$e_{t+1}(k) = \eta_{t+1}(k) \prod_{i \in \mathcal{S}_t} \mathcal{N}(r_{i, t+1}; \mu_k, s_k^2)$$

as the emission and run forward–backward over everything. In log terms, the product over roughly 500 stocks outweighs the single market term by a factor of hundreds. The cross-section would then decide what the regimes are, and their meaning as *market* regimes would be lost. The base case avoids this by design.

Numerics. Compute

$$\log \sum_k \exp \left(\log \pi_k(t) + \sum_{i \in \mathcal{S}_t} \log \mathcal{N}_{ik} \right)$$

with a single `logsumexp`. Never multiply densities directly.

7.3 The objective used: the per-row composite likelihood (Q24, decided)

The study does *not* maximise ℓ_{date} . It maximises

$$\ell_{\text{row}}(\psi, s) = \sum_{t=1}^{T-1} \sum_{i \in \mathcal{S}_t} \log \sum_{k=1}^K \pi_k(t) \mathcal{N}(r_{i,t+1}; \mu_k(x_{i,t}), s_k^2) \quad (80)$$

(decided 2026-10-05, Q24). This section gives the short version; Appendix D gives the full theory, step by step. There are two ways to read this objective.

Reading 1: the likelihood of a different model. If every stock-day drew its own regime from the day’s prior, (80) would be that model’s exact likelihood. In the graph of Section 6 the latent node would move inside the stock plate.

Reading 2: a composite likelihood of *our* model. Keep the model of Section 6, with one regime per day, and ask for the distribution of a single stock’s return on its own. Integrating out all other stocks (each integrates to one given z_{t+1}) and then z_{t+1} gives

$$p(r_{i,t+1} \mid x_{i,t}, \mathcal{F}_t) = \sum_{k=1}^K \pi_k(t) \mathcal{N}(r_{i,t+1}; \mu_k(x_{i,t}), s_k^2). \quad (81)$$

This is exactly the term inside (80). So ℓ_{row} is the sum of the *correct* marginal log-likelihoods of the per-date model: it keeps every stock’s own distribution and discards only the dependence between stocks on the same date. Such an objective is a *composite likelihood*, in this case the *independence likelihood* (Lindsay 1988; Varin, Reid & Firth 2011).

Why it is a valid estimator. Each term is a true log-density, so its score has expectation zero at the true parameters. A sum of unbiased score terms is an unbiased estimating equation, and under the usual regularity conditions its maximiser is consistent. The cost is efficiency, because the joint information carried by the co-movement of stocks on a date is not used, and its curvature does not give valid standard errors. Neither matters here: inference in the study comes from walk-forward rank ICs, not from the likelihood. Composite likelihood is used in finance for exactly this kind of large cross-section, where the full likelihood is unreliable (Pakel, Shephard, Sheppard & Engle 2021).

Why it is preferred to ℓ_{date} .

1. **Robustness to A3.** ℓ_{date} uses its extra information only through A3(a): given the regime, the stocks are independent. That assumption is false, because industry peers share shocks after the market is removed, and the market-neutral target leaves residual beta exposure (Q25). ℓ_{date} then counts correlated stocks as independent evidence about the regime. With m stocks per cluster correlated at ρ , the cross-section carries the information of only

$$N_{\text{eff}} \approx \frac{N}{1 + (m-1)\rho}$$

independent stocks (Kish’s design effect). For example, $N = 500$, $m = 45$ and $\rho = 0.1$ give $N_{\text{eff}} \approx 93$, so the per-date posterior would be overconfident by a factor of about five. ℓ_{row} needs only the marginals to be right, so this violation does not bias it (Liang & Zeger 1986).

2. **The gate stays in charge.** Under ℓ_{date} the responsibility $\gamma_t(k)$ is almost always 0 or 1 and is set by the day’s returns (see the sharpness calculation below), so the cross-section re-decides the regime every day. Under ℓ_{row} one stock carries little evidence, so $\gamma_{i,t}(k)$ stays close to $\pi_k(t)$, and each expert learns mainly from the days that the *gate* assigns to its regime. That is the point of freezing the gate (Q19): the regimes remain *market* regimes.
3. **Practicalities.** It is what the code already implements (`losses.mixture_nll`, a log-sum-exp per row); mini-batches may be any rows; training is smoother.

Its weakness, and the diagnostic. The experts specialise less sharply, and the expert with the larger s_k can drift into absorbing outlying stock-days instead of representing a regime. After training, check whether each expert’s share of the training weight follows the gate’s $\pi_k(t)$ or the size of the residuals.

7.4 How Stage 2 is maximised

The EM view. The E-step gives each stock-day’s posterior over the regime after seeing its own return:

$$\gamma_{i,t}(k) = \frac{\pi_k(t) \mathcal{N}(r_{i,t+1}; \mu_k(x_{i,t}), s_k^2)}{\sum_j \pi_j(t) \mathcal{N}(r_{i,t+1}; \mu_j(x_{i,t}), s_j^2)}. \quad (82)$$

In the M-step each expert solves a weighted least-squares problem, with weight $\gamma_{i,t}(k)$ on row (i, t) , and the noise has a closed form:

$$\min_{\psi_k} \sum_t \sum_{i \in \mathcal{S}_t} \gamma_{i,t}(k) (r_{i,t+1} - \mu_k(x_{i,t}; \psi_k))^2, \quad s_k^2 = \frac{\sum_t \sum_i \gamma_{i,t}(k) (r_{i,t+1} - \mu_k(x_{i,t}))^2}{\sum_t \sum_i \gamma_{i,t}(k)}. \quad (83)$$

With linear experts the first problem has a closed form (Appendix A(i)); with MLPs it does not.

What is actually done: gradient ascent on ℓ_{row} . By Fisher’s identity (Section 4.5) the gradients carry the same weights:

$$\frac{\partial \ell_{\text{row}}}{\partial \mu_k(x_{i,t})} = \gamma_{i,t}(k) \frac{r_{i,t+1} - \mu_k(x_{i,t})}{s_k^2}, \quad \frac{\partial \ell_{\text{row}}}{\partial \log s_k} = \sum_t \sum_i \gamma_{i,t}(k) \left[\frac{(r_{i,t+1} - \mu_k)^2}{s_k^2} - 1 \right]. \quad (84)$$

Each step recomputes γ with the current parameters and moves a little on the weighted problem. Backpropagation (Section 4.6) carries the first gradient into every expert’s weights. Mini-batches may contain any rows.

Numerics. For each row compute $\log \sum_k \exp(\log \pi_k(t) + \log \mathcal{N}_{ik})$ with a single `logsumexp`.

7.5 For comparison: how sharp the per-date posterior is

Under ℓ_{date} the E-step pools the whole cross-section:

$$\log \frac{\gamma_t(k)}{\gamma_t(j)} = \log \frac{\pi_k(t)}{\pi_j(t)} + \sum_{i \in \mathcal{S}_t} \log \frac{\mathcal{N}(r_{i,t+1}; \mu_k, s_k^2)}{\mathcal{N}(r_{i,t+1}; \mu_j, s_j^2)}. \quad (85)$$

If regime k is true, each term of the sum has expectation equal to a Kullback–Leibler divergence, which is positive, so the sum grows roughly in proportion to N_t . For example, two experts that differ only in noise level (1.5% against 1%) give

$$\text{KL}(\mathcal{N}(0, 1.5^2) \parallel \mathcal{N}(0, 1^2)) = \log \frac{1}{1.5} + \frac{1.5^2}{2} - \frac{1}{2} \approx 0.22 \text{ nats per stock}, \quad (86)$$

so about 110 nats on a day with 500 stocks. The per-date γ is therefore nearly always 0 or 1, whereas the per-row γ stays close to the prior.

	Per date ℓ_{date}	Per row ℓ_{row} (used)
Relation to Section 6	exact likelihood	composite (independence) likelihood
Needs A3(a)	yes	no, only correct marginals
Responsibilities	near 0/1, set by the day’s returns	soft, close to the gate’s π
Experts specialise by	days whose cross-section fits	the gate’s regimes (risk: tail rows)
Mini-batches	whole dates	any rows
Forecast formula	$\sum_k \pi_k \mu_k$	identical

The forecast formula is the same under both, because forecasts use π and never γ ; only the training of the experts differs.

7.6 What freezing the gate changes

- No gradient reaches the gate. The $\pi_k(t)$ are fixed numbers, and the experts adapt to them.
- The HMM fixes the labels: expert k is the expert for HMM state k , in the order chosen in Section 5.6.
- On a given day the experts can still “disagree” with the gate, because γ uses the returns and π does not. A scatter of π against γ (Weigend et al. 1995) shows how informative the gate is.
- Freezing causes no look-ahead: Baum–Welch runs on the training block only, and only the forward pass runs after it.

8 Prediction and posteriors (base case)

8.1 The information at forecast time

At the close of day t the forecast may use $\mathcal{G}_t$: the market history (through the forward pass only), today's characteristics $x_{i,t}$, and the frozen parameters. The gate is the $\mathcal{F}_t$ -measurable probability

$$\pi_k(t) = \mathbb{P}(z_{t+1} = k \mid \mathcal{F}_t) = \sum_j A_{jk} \xi_{t|t}(j). \quad (87)$$

8.2 The predictive distribution of one stock

Apply the sum rule over z_{t+1} :

$$p(r_{i,t+1} \mid x_{i,t}, \mathcal{F}_t) = \sum_{k=1}^K \pi_k(t) \mathcal{N}(r_{i,t+1}; \mu_k(x_{i,t}), s_k^2). \quad (88)$$

This is a Gaussian mixture, so it can be skewed and fat-tailed even though every expert is Gaussian. It is the full forecast; everything below summarises it.

8.3 The point forecast

By the law of total expectation,

$$\hat{r}_{i,t+1} = \mathbb{E}[r_{i,t+1} \mid x_{i,t}, \mathcal{F}_t] = \sum_k \pi_k(t) \mu_k(x_{i,t}) = f_0(x_{i,t}) + \sum_k \pi_k(t) c_k(x_{i,t}). \quad (89)$$

The last step uses $\sum_k \pi_k(t) = 1$: the base comes out of the sum, so the forecast is the base plus the gate-weighted corrections. The rank IC is computed from $\hat{r}_{i,t+1}$.

8.4 The predictive variance

The second moment is $\mathbb{E}[r^2 \mid \cdot] = \sum_k \pi_k(t) (s_k^2 + \mu_k^2)$. Subtracting $\hat{r}^2$ and rearranging gives the law of total variance:

$$\text{Var}(r_{i,t+1} \mid x_{i,t}, \mathcal{F}_t) = \underbrace{\sum_k \pi_k(t) s_k^2}_{\text{noise within a regime}} + \underbrace{\sum_k \pi_k(t) (\mu_k(x_{i,t}) - \hat{r}_{i,t+1})^2}_{\text{experts disagree, regime uncertain}}. \quad (90)$$

The first term remains even if tomorrow's regime were known. The second exists only because the regime is unknown and the experts predict different things; it vanishes if the gate is certain or if the experts agree for this stock.

8.5 Covariance between stocks (per-date model)

Two stocks share z_{t+1} and are independent given it, so uncertainty about the regime makes them co-move:

$$\text{Cov}(r_{i,t+1}, r_{j,t+1} \mid \cdot) = \sum_k \pi_k(t) \mu_k(x_{i,t}) \mu_k(x_{j,t}) - \hat{r}_{i,t+1} \hat{r}_{j,t+1} = \sum_k \pi_k(t) (\mu_k(x_{i,t}) - \hat{r}_{i,t+1}) (\mu_k(x_{j,t}) - \hat{r}_{j,t+1}). \quad (91)$$

This covariance is a property of the model; it is not affected by training with the per-row objective (Q24), which only discards it from the estimation. Under the alternative per-row *model* it would be zero. It matters only for portfolios built from the predictive distribution; ranking uses $\hat{r}$ alone.

8.6 The study's horizon: $h = 5$

The study uses a daily gate, a daily cross-section and a 5-day target (Q21, decided 2026-10-05). The derivations in these notes are written for one step, with $r_{i,t+1}$; in the study that step is the 5-day window $(t, t + 5]$, and $r_{i,t+1}$ stands for the market-neutral return over it. The regime can change inside the window, and the regime forecast j days ahead is $\mathbb{P}(z_{t+j} = k \mid \mathcal{F}_t) = [\xi_{t|t}^\top A^j]_k$. The likelihood keeps a single regime for the whole window, which is a reasonable approximation because 5 days is short against typical regime durations of 20–50 days (Section 5.3); the window-average weight below makes the point forecast exact even when the regime changes inside the window. The gate weight for the window is the average of the $j = 1, \dots, 5$ forecasts (decided 2026-10-05, Q27; Section 8.9). Consecutive daily targets overlap by four days: statistics are overlap-corrected (Hansen–Hodrick), and the training likelihood is used for estimation, not inference.

8.7 Every regime probability, and where it may be used

Quantity	Measurable w.r.t.	Computed from	Used for	At fore- cast time
filtered $\xi_{t t}$	$\mathcal{F}_t$	forward pass ($\hat{\alpha}_t$)	building the gate	yes
gate $\bar{w}_k(t)$ ($= \pi_k(t)$ if $h = 1$)	$\mathcal{F}_t$	$\xi_{t t}$ and A	weighting the experts	yes
smoothed (gate) γ_t	$\mathcal{F}_T$	$\hat{\alpha}_t \hat{\beta}_t$	Baum–Welch M-step, training block	no
pairwise $\xi_t(j, k)$	$\mathcal{F}_T$	$\hat{\alpha}, A, \eta, \hat{\beta}$	updating A , training block	no
responsibility $\gamma_{i,t}(k)$	$\mathcal{G}_{t+1}$	Bayes per stock-day, uses $r_{i,t+1}$	weighting expert gradients in training	no

8.8 Why smoothed probabilities leak the future

$$\gamma_t(k) \propto \alpha_t(k) \beta_t(k), \quad \beta_t(k) = p(m_{t+1}, \dots, m_T \mid z_t = k). \quad (92)$$

Suppose the market crashes on day $t + 3$. That crash is far more likely under the stressed regime, so β_t shifts weight to the stressed regime on day t , three days before the crash, when no past data pointed to it. A gate built on smoothed probabilities therefore enters crashes in advance: the backtest looks excellent and means nothing. In the language of Section 1.4, the smoothed probability is not $\mathcal{F}_t$ -measurable, so a forecast built on it is not adapted.

Leaks that are easy to miss.

- Refitting Baum–Welch on data that include validation or test dates. The M-step uses smoothed posteriors, so future data enter θ_g .
- Training on smoothed weights and testing on filtered ones. This is not look-ahead at test time, but the experts face a different kind of input than they were trained on. That is why the filtered series is used on training dates too.
- Using the responsibilities γ as features at forecast time. They contain the target itself.
- The purge gap (Q22, open). The forward pass must be carried continuously through the dates removed between training and test.

8.9 The gate weight for an h -day target (Q27, decided)

Three $\mathcal{F}_t$ -measurable candidates are available at the close of day t , all built from the filtered probability:

- the *filtered* probability $\xi_{t|t}(k)$, today's regime (used by `Model_Derivation_BaseCase.md` before this decision, and by the code's Hamilton gate);
- the *one-step predicted* probability $\pi_k(t) = [\xi_{t|t}^\top A]_k$, tomorrow's regime (Section 6 with $h = 1$);
- the *window average*, the expected share of the window $(t, t + h]$ spent in regime k :

$$\bar{w}_k(t) = \frac{1}{h} \sum_{j=1}^h [\xi_{t|t}^\top A^j]_k. \quad (93)$$

Why the window average is exact for the point forecast. The target is the sum of h daily returns. Suppose each day's expected return is set by that day's regime, with regime k contributing $\mu_k(x_{i,t})/h$ per day. By the law of total expectation, day by day,

$$\mathbb{E}[y_{i,t} \mid \mathcal{F}_t] = \sum_{j=1}^h \sum_k \mathbb{P}(z_{t+j} = k \mid \mathcal{F}_t) \frac{\mu_k(x_{i,t})}{h} = \sum_k \bar{w}_k(t) \mu_k(x_{i,t}). \quad (94)$$

So the gate-weighted forecast with (93) is exactly the expected h -day return. The expected within-regime noise also adds up over days and is matched in the same way. The filtered weight assumes the regime never changes during the window, and the one-step weight assumes it changes at most once. For $h = 1$, (93) reduces to the one-step predicted probability, so every one-step formula in these notes is the $h = 1$ case of this rule.

How large the differences are. With the example A of Section 5.3 and a filtered belief of certain stress, the weight on stress is 1.00 (filtered), 0.95 (one-step) and 0.86 (window average with $h = 5$). A descriptive fit of a two-state Hamilton model to the CRSP S&P 500 index, 2015–2024, gives an average gap of 0.055 between the filtered and window-average weights on the stressed regime, with a maximum of 0.10. The gap is largest when the filter is sure of stress, because stress tends to end.

Decision. The window average is used on training and test dates alike (Q27, decided 2026-10-05). Costs accepted: slightly softer weights, and dependence on a well-estimated A , whose errors compound over h steps. The likelihood with these weights remains the one-regime-per-window approximation: the exact distribution of an h -day return is a mixture over regime *paths*.

9 Assumptions and design checklist (base case)

9.1 Modelling assumptions

	Assumption	Comment
A0	Characteristics carry no information about the regime path	keeps stock information out of the gate
A1	First-order Markov regimes with one A per training block	geometric durations; A re-estimated in every walk-forward fold
A2	The market return depends only on its own day's regime (Gaussian, switching mean and variance)	the AR variant breaks it on purpose (G-2)
A3	Given z_{t+1} and $x_{i,t}$, the stock returns are independent of each other, of m_{t+1} and of other days	weak for raw returns (common market shock); reasonable for the market-neutral target used here; industry peers remain correlated, which the per-row objective tolerates (Q24)
A4	One noise level s_k per regime, the same for all stocks	no stock-specific volatility in the experts
A5	All parameters constant within a training block	re-estimated per fold

9.2 Design choices and their status

“Decided” means decided for the base case or recorded as resolved in `Advisor_Questions.md`.

“Open” gives the question number.

Item	Base-case choice or options	Status
Gate fitted separately and frozen	Baum–Welch on the training block, forward pass afterwards	decided (Q19)
Base pre-trained and frozen	f_0 fitted first, never updated	decided (Q7)
Filtered, never smoothed, as the basis of the gate	on training and test dates alike	decided (base-case derivation)
Gate weight for the h -day target	window average $\bar{w}_k(t)$ of the 1- to h -step-ahead probabilities (Section 8.9)	decided (Q27)
Number of regimes K	e.g. 2 or 3	open (Q18)
Gate series m_t	CRSP value-weighted S&P 500 index, daily log return; filter runs through the purge gap	decided (Q23, Q22)
Frequency and horizon h	daily gate, daily cross-section, $h = 5$ (overlapping targets)	decided (Q21)
Filter through the purge gap	run forward continuously or restart	open (Q22)
Likelihood for the experts	ℓ_{row} , the composite likelihood of the per-date model (Section 7.3)	decided (Q24)
Objective	base case: mixture log-likelihood only	decided here; Q20 open for the thesis
State labelling across folds	order states, e.g. by σ_k	to check in code
Residual form	$\mu_k = f_0 + c_k$, final layer of c_k zero-initialised	implemented; Q6 and Q14 open
Expert noise s_k	learned scalar per expert, frozen for the first 100 steps	implemented; no floor or prior in the base case
Regime information	enters only through the gate; experts see $x_{i,t}$ only	decided
Target	market-neutral return: forward return minus the date’s equal-weighted cross-sectional mean	decided (Q25); not yet implemented
Expert and base architecture	MLP widths and depth	open (Q8; depth grid per Q11)
Training window and weighting	full block vs decay; crash preservation	open (Q16, B-2)
Hidden-layer initialisation	switch X-1	open
Mini-batching	any rows (under ℓ_{row})	decided (Q24)

9.3 The base case in one paragraph

A K -state Gaussian HMM is fitted by Baum–Welch to the daily market series on the training block and frozen. Its forward pass gives, at the close of each day t , the filtered probability $\xi_{t|t}$; the gate weight for the 5-day target is its window average $\bar{w}_k(t)$ (93), which equals $\pi_k(t) = \mathbb{P}(z_{t+1} = k \mid \mathcal{F}_t)$ when $h = 1$ (Q27). A frozen base MLP f_0 and K zero-initialised MLP corrections c_k define expert means $\mu_k = f_0 + c_k$ with noise levels s_k . The corrections and noise levels are fitted by gradient ascent on the per-row (composite) mixture log-likelihood of tomorrow’s returns with the gate held fixed (Q24). The forecast is $\hat{r}_{i,t+1} = f_0(x_{i,t}) + \sum_k \pi_k(t) c_k(x_{i,t})$, and its full predictive distribution is the mixture (88).

A Solution to Exercise A1

Throughout, $\mathcal{N}_{nk} := \mathcal{N}(y_n; \beta_k^\top x_n, s_k^2)$ and $\pi_{nk} := \pi_k(x_n; w)$.

(a) Generative story and graph. For each n : given x_n , draw the expert label, then the target,

$$z_n \sim \text{Cat}(\pi_{n1}, \dots, \pi_{nK}), \quad y_n \sim \mathcal{N}(\beta_{z_n}^\top x_n, s_{z_n}^2).$$

The graph is the one in Section 4.2 with $u_n = x_n$. Observed: x_n, y_n . Latent: z_n . Parameters: w (entering at z_n) and β_k, s_k (entering at y_n). The plate is over n , so the z_n are independent across observations.

(b) Complete-data likelihood. Exactly one $z_{nk} = 1$ for each n , so

$$p(Y, Z \mid X; \Psi) = \prod_{n=1}^N \prod_{k=1}^K [\pi_{nk} \mathcal{N}_{nk}]^{z_{nk}}, \quad \log L_c(\Psi) = \sum_{n=1}^N \sum_{k=1}^K z_{nk} [\log \pi_{nk} + \log \mathcal{N}_{nk}]. \quad (95)$$

(c) Observed-data log-likelihood. Summing over Z means summing over every z_n . The factor for observation n depends only on z_n , so the sum moves inside the product. This is what fails in an HMM, where transitions link neighbouring latents:

$$p(Y \mid X; \Psi) = \sum_{z_1} \cdots \sum_{z_N} \prod_n \prod_k [\pi_{nk} \mathcal{N}_{nk}]^{z_{nk}} = \prod_n \sum_{z_n} \prod_k [\pi_{nk} \mathcal{N}_{nk}]^{z_{nk}} = \prod_n \sum_k \pi_{nk} \mathcal{N}_{nk}. \quad (96)$$

The last step holds because, when z_n has its 1 in position j , every factor with $k \neq j$ equals 1. Hence

$$\log L(\Psi) = \sum_{n=1}^N \log \sum_{k=1}^K \pi_k(x_n; w) \mathcal{N}(y_n; \beta_k^\top x_n, s_k^2). \quad (97)$$

The sum inside the logarithm couples w , every β_k and every s_k , and the function is not concave. Setting the gradient to zero gives equations in which the responsibilities themselves depend on all the parameters, so there is no closed-form solution.

(d) E-step. By Bayes' rule,

$$\tau_{nk}^{(q)} = p(z_{nk} = 1 \mid y_n, x_n; \Psi^{(q)}) = \frac{p(z_{nk} = 1 \mid x_n) p(y_n \mid x_n, z_{nk} = 1)}{p(y_n \mid x_n)} = \frac{\pi_{nk}^{(q)} \mathcal{N}_{nk}^{(q)}}{\sum_j \pi_{nj}^{(q)} \mathcal{N}_{nj}^{(q)}}. \quad (98)$$

(e) The Q -function. $\log L_c$ is linear in z_{nk} , so its expectation replaces z_{nk} by $\tau_{nk}^{(q)}$:

$$Q(\Psi; \Psi^{(q)}) = \underbrace{\sum_n \sum_k \tau_{nk}^{(q)} \log \pi_k(x_n; w)}_{Q_{\text{gate}}(w)} + \underbrace{\sum_{k=1}^K \sum_n \tau_{nk}^{(q)} \log \mathcal{N}(y_n; \beta_k^\top x_n, s_k^2)}_{Q_k(\beta_k, s_k^2)}. \quad (99)$$

The gate term involves only w , and each Q_k involves only (β_k, s_k^2) , so the M-step splits into $K+1$ separate problems.

(f) **Expert M-step.** Writing out the Gaussian,

$$Q_k = \sum_n \tau_{nk} \left[-\frac{1}{2} \log 2\pi - \frac{1}{2} \log s_k^2 - \frac{(y_n - \beta_k^\top x_n)^2}{2s_k^2} \right]. \quad (100)$$

Differentiate with respect to β_k and set to zero:

$$\frac{\partial Q_k}{\partial \beta_k} = \frac{1}{s_k^2} \sum_n \tau_{nk} (y_n - \beta_k^\top x_n) x_n = 0 \quad \Rightarrow \quad \left(\sum_n \tau_{nk} x_n x_n^\top \right) \beta_k = \sum_n \tau_{nk} x_n y_n. \quad (101)$$

With X the $N \times d$ design matrix, y the target vector and $W_k = \text{diag}(\tau_{1k}, \dots, \tau_{Nk})$,

$$\beta_k^{(q+1)} = (X^\top W_k X)^{-1} X^\top W_k y. \quad (102)$$

This is weighted least squares (Chamroukhi & Nguyen eq. 18). Next differentiate with respect to s_k^2 :

$$\frac{\partial Q_k}{\partial s_k^2} = -\frac{1}{2s_k^2} \sum_n \tau_{nk} + \frac{1}{2s_k^4} \sum_n \tau_{nk} (y_n - \beta_k^\top x_n)^2 = 0 \quad \Rightarrow \quad s_k^{2(q+1)} = \frac{\sum_n \tau_{nk} (y_n - \beta_k^{(q+1)\top} x_n)^2}{\sum_n \tau_{nk}}. \quad (103)$$

(g) **Gate M-step.** Using $\log \pi_{nk} = w_k^\top x_n - \log \sum_j \exp(w_j^\top x_n)$ and $\sum_k \tau_{nk} = 1$,

$$Q_{\text{gate}}(w) = \sum_n \left[\sum_k \tau_{nk} w_k^\top x_n - \log \sum_j \exp(w_j^\top x_n) \right]. \quad (104)$$

Gradient, for $k = 1, \dots, K-1$:

$$\frac{\partial Q_{\text{gate}}}{\partial w_k} = \sum_n (\tau_{nk} - \pi_{nk}) x_n. \quad (105)$$

Hessian blocks, from $\partial \pi_{nk} / \partial w_l = \pi_{nk} (\delta_{kl} - \pi_{nl}) x_n$:

$$\frac{\partial^2 Q_{\text{gate}}}{\partial w_k \partial w_l^\top} = - \sum_n \pi_{nk} (\delta_{kl} - \pi_{nl}) x_n x_n^\top. \quad (106)$$

Concavity. Stack $w = (w_1, \dots, w_{K-1})$. The Hessian is $-\sum_n (\text{diag}(\pi_n) - \pi_n \pi_n^\top) \otimes x_n x_n^\top$. Here π_n means the first $K-1$ components. The matrix $\text{diag}(\pi_n) - \pi_n \pi_n^\top$ is the covariance matrix of those components of a one-hot categorical vector, so it is positive semidefinite, and a Kronecker product of positive semidefinite matrices is positive semidefinite. The Hessian is therefore negative semidefinite and Q_{gate} is concave: Newton's method finds its global maximum.

No closed form. The first-order condition $\sum_n (\tau_{nk} - \pi_{nk}(w)) x_n = 0$ is nonlinear in w , because w sits inside exponentials.

One Newton-Raphson step:

$$w^{\text{new}} = w^{\text{old}} - H^{-1} g. \quad (107)$$

For $K = 2$ this is the familiar IRLS form. With $R = \text{diag}(\pi_{n1}(1 - \pi_{n1}))$ and $\tau = (\tau_{n1})_n$,

$$w_1^{\text{new}} = w_1^{\text{old}} + (X^\top R X)^{-1} X^\top (\tau - \pi) = (X^\top R X)^{-1} X^\top R \zeta, \quad \zeta = X w_1^{\text{old}} + R^{-1}(\tau - \pi). \quad (108)$$

This is a weighted least-squares regression on a working response ζ , repeated until convergence (Bishop §4.3.3; Chamroukhi & Nguyen §4.3.2).

(h) Gate without input. Now $\pi_k = \alpha_k$ is a constant. Maximise $\sum_n \sum_k \tau_{nk} \log \alpha_k$ subject to $\sum_k \alpha_k = 1$:

$$\frac{\partial}{\partial \alpha_k} \left[\sum_{n,k} \tau_{nk} \log \alpha_k + \lambda \left(1 - \sum_k \alpha_k \right) \right] = \frac{\sum_n \tau_{nk}}{\alpha_k} - \lambda = 0. \quad (109)$$

Summing $\alpha_k = \sum_n \tau_{nk} / \lambda$ over k gives $\lambda = N$, so $\alpha_k^{(q+1)} = \frac{1}{N} \sum_n \tau_{nk}^{(q)}$ (Chamroukhi & Nguyen eq. 8). Equivalently, the gradient in (g) with $x_n \equiv 1$ is $\sum_n (\tau_{nk} - \alpha_k) = 0$.

(i) The panel version. One z_t per date, shared by all $i \in \mathcal{S}_t$; gate $\pi_k(u_t; w)$; write $\mathcal{N}_{itk} = \mathcal{N}(r_{i,t+1}; \beta_k^\top x_{i,t}, s_k^2)$.

$$(b) \quad p(R, Z \mid X, U) = \prod_t \prod_k \left[\pi_k(u_t) \prod_{i \in \mathcal{S}_t} \mathcal{N}_{itk} \right]^{z_{tk}}, \quad (110)$$

$$(c) \quad \log L = \sum_t \log \sum_k \pi_k(u_t) \prod_{i \in \mathcal{S}_t} \mathcal{N}_{itk}, \quad (111)$$

$$(d) \quad \tau_{tk} = \frac{\pi_k(u_t) \prod_i \mathcal{N}_{itk}}{\sum_j \pi_j(u_t) \prod_i \mathcal{N}_{itj}}, \quad (112)$$

$$(f) \quad \beta_k = \left(\sum_t \tau_{tk} X_t^\top X_t \right)^{-1} \sum_t \tau_{tk} X_t^\top r_{t+1}, \quad s_k^2 = \frac{\sum_t \tau_{tk} \|r_{t+1} - X_t \beta_k\|^2}{\sum_t \tau_{tk} N_t}, \quad (113)$$

where X_t is the $N_t \times d$ matrix of today's characteristics and r_{t+1} the vector of tomorrow's returns. These are Chamroukhi & Nguyen eqs. 18–19 with curve = date and point = stock.

What changes in τ . The log-odds are

$$\log \frac{\tau_{tk}}{\tau_{tj}} = \log \frac{\pi_k(u_t)}{\pi_j(u_t)} + \sum_{i \in \mathcal{S}_t} \log \frac{\mathcal{N}_{itk}}{\mathcal{N}_{itj}}. \quad (114)$$

The evidence term is a sum of N_t terms whose expectation under the true regime is N_t times a positive Kullback–Leibler divergence. It grows linearly in N_t , so with hundreds of stocks the posterior is close to 0 or 1 and the prior hardly matters. In the per-row model only one term enters (Section 7.5 and Appendix D).

(j) EM never decreases the likelihood. Since $\log p(Y \mid \Psi) = \log p(Y, Z \mid \Psi) - \log p(Z \mid Y, \Psi)$ for every Z , take the expectation over $Z \sim p(\cdot \mid Y; \Psi^{(q)})$:

$$\log L(\Psi) = \underbrace{\mathbb{E}[\log p(Y, Z \mid \Psi)]}_{Q(\Psi; \Psi^{(q)})} - \underbrace{\mathbb{E}[\log p(Z \mid Y, \Psi)]}_{H(\Psi; \Psi^{(q)})}. \quad (115)$$

$H(\Psi; \Psi^{(q)})$ is a cross-entropy between $p(\cdot \mid Y; \Psi^{(q)})$ and $p(\cdot \mid Y; \Psi)$. By Gibbs' inequality it is minimised at $\Psi = \Psi^{(q)}$, so $H(\Psi^{(q+1)}; \Psi^{(q)}) \geq H(\Psi^{(q)}; \Psi^{(q)})$. Therefore

$$\log L(\Psi^{(q+1)}) - \log L(\Psi^{(q)}) = \underbrace{[Q(\Psi^{(q+1)}; \Psi^{(q)}) - Q(\Psi^{(q)}; \Psi^{(q)})]}_{\geq 0 \text{ by the M-step}} + \underbrace{[H(\Psi^{(q+1)}; \Psi^{(q)}) - H(\Psi^{(q)}; \Psi^{(q)})]}_{\geq 0 \text{ by Gibbs}} \geq 0. \quad (116)$$

B Solution to Exercise A2

Write $\eta_t(k) = \mathcal{N}(m_t; \nu_k, \sigma_k^2)$, $\mathcal{N}_{itk} = \mathcal{N}(r_{i,t+1}; \beta_k^\top x_{i,t}, s_k^2)$, X_t for the $N_t \times d$ matrix of today's characteristics and r_{t+1} for the vector of tomorrow's returns.

(a) Graph. The graph is the one in Section 6.2, with linear expert means. Observed: $m_{1:T}$ and $r_{i,t+1}$. Conditioned on only: $x_{i,t}$ and $\mathcal{S}_t$. Latent: $z_{1:T}$. Parameters: ρ, A, ν, σ (gate) and β_k, s_k (experts).

(b) Joint. This is Section 6.3 with assumptions A0–A3 and $\mu_k(x) = \beta_k^\top x$:

$$p(z_{1:T}, m_{1:T}, R \mid X) = \rho_{z_1} \prod_{t=1}^{T-1} A_{z_t z_{t+1}} \prod_{t=1}^T \eta_t(z_t) \prod_{t=1}^{T-1} \prod_{i \in \mathcal{S}_t} \mathcal{N}_{i,t,z_{t+1}}. \quad (117)$$

(c) Emissions and the forward recursion. Group everything that day $t+1$ emits:

$$e_1(k) = \eta_1(k), \quad e_{t+1}(k) = \eta_{t+1}(k) \prod_{i \in \mathcal{S}_t} \mathcal{N}_{itk}. \quad (118)$$

Then the model is a standard HMM with emissions e_t , and

$$L = \sum_{z_1, \dots, z_T} \rho_{z_1} e_1(z_1) \prod_{t=1}^{T-1} A_{z_t z_{t+1}} e_{t+1}(z_{t+1}). \quad (119)$$

The scaled forward recursion (Bishop §13.2.4) is

$$c_1 = \sum_k \rho_k e_1(k), \quad \hat{\alpha}_1(k) = \frac{\rho_k e_1(k)}{c_1}, \quad c_{t+1} = \sum_k e_{t+1}(k) \sum_j \hat{\alpha}_t(j) A_{jk}, \quad \hat{\alpha}_{t+1}(k) = \frac{e_{t+1}(k) \sum_j \hat{\alpha}_t(j) A_{jk}}{c_{t+1}}, \quad (120)$$

with $\log L = \sum_t \log c_t$. Each step costs $O(K^2)$ for the sum over j , plus $O(N_t K)$ to evaluate the emissions, so the whole pass costs $O(TK^2 + K \sum_t N_t)$ instead of the K^T terms of the path sum.

(d) Backward recursion and posteriors.

$$\hat{\beta}_T(k) = 1, \quad \hat{\beta}_t(j) = \frac{\sum_k A_{jk} e_{t+1}(k) \hat{\beta}_{t+1}(k)}{c_{t+1}}, \quad (121)$$

$$\gamma_t(k) = \hat{\alpha}_t(k) \hat{\beta}_t(k), \quad \xi_t(j, k) = \frac{\hat{\alpha}_t(j) A_{jk} e_{t+1}(k) \hat{\beta}_{t+1}(k)}{c_{t+1}}. \quad (122)$$

(e) Q and the M-step. Taking the expectation of the log of (b) under the posterior,

$$Q = \sum_k \gamma_1(k) \log \rho_k + \sum_{t=1}^{T-1} \sum_{j,k} \xi_t(j, k) \log A_{jk} + \sum_{t=1}^T \sum_k \gamma_t(k) \log \eta_t(k) + \sum_{t=1}^{T-1} \sum_k \gamma_{t+1}(k) \sum_{i \in \mathcal{S}_t} \log \mathcal{N}_{itk}. \quad (123)$$

Each block is maximised separately:

$$\rho_k = \gamma_1(k), \quad A_{jk} = \frac{\sum_{t=1}^{T-1} \xi_t(j, k)}{\sum_{t=1}^{T-1} \gamma_t(j)}, \quad \nu_k = \frac{\sum_t \gamma_t(k) m_t}{\sum_t \gamma_t(k)}, \quad \sigma_k^2 = \frac{\sum_t \gamma_t(k) (m_t - \nu_k)^2}{\sum_t \gamma_t(k)}, \quad (124)$$

$$\beta_k = \left(\sum_{t=1}^{T-1} \gamma_{t+1}(k) X_t^\top X_t \right)^{-1} \sum_{t=1}^{T-1} \gamma_{t+1}(k) X_t^\top r_{t+1}, \quad s_k^2 = \frac{\sum_t \gamma_{t+1}(k) \|r_{t+1} - X_t \beta_k\|^2}{\sum_t \gamma_{t+1}(k) N_t}. \quad (125)$$

Comparison with Chamroukhi & Nguyen eq. (46), $\beta_{kr} = [\sum_i \tau_{ik} X_i^\top W_{ikr} X_i]^{-1} \sum_i \tau_{ik} X_i^\top W_{ikr} y_i$:

- there is one curve and one cluster, so $\tau = 1$;
- their diagonal weight matrix W_{ikr} holds one regime posterior per time point;
- in our model each time point is a whole date, so the weight $\gamma_{t+1}(k)$ is repeated across the N_t stocks of that date and multiplies the block $X_t^\top X_t$.

(f) The two-stage version.

- *Stage 1*: Baum–Welch as above but with $e_t = \eta_t$, the market only (Section 5.5).
- *Stage 2*: the gate $\pi_k(t) = \sum_j A_{jk} \xi_{t|t}(j)$ comes from the market-only filter.
 - E-step: $\gamma_t^{\text{cs}}(k) \propto \pi_k(t) \prod_i \mathcal{N}_{itk}$, per date, with no recursion.
 - M-step: the same β_k and s_k^2 formulas as in (e), with $\gamma_t^{\text{cs}}(k)$ in place of $\gamma_{t+1}(k)$.

The joint fit uses two things that the two-stage fit ignores:

1. the cross-section’s evidence about the regime, carried *across dates* by A in both directions (forward and backward);
2. that evidence when re-estimating A , ν and σ .

Because $\prod_i \mathcal{N}_{itk}$ dominates η_t , the joint fit lets the cross-section define the regimes. The two-stage fit keeps them market-defined.

(g) Forecasting.

- *Two-stage*: at the close of t , use $\pi_k(t) = \mathbb{P}(z_{t+1} = k \mid \mathcal{F}_t)$.
- *Joint model*: use $\sum_j A_{jk} \hat{\alpha}_t(j)$ from the joint forward pass. This conditions on past market data *and* past returns, so it is $\mathcal{G}_t$ -measurable and legal.

The smoothed $\gamma_{t+1} = \hat{\alpha}_{t+1} \hat{\beta}_{t+1}$ is look-ahead twice over: $\hat{\alpha}_{t+1}$ already contains e_{t+1} , i.e. tomorrow’s returns, which are the target; and $\hat{\beta}_{t+1}$ contains every later observation. It is not $\mathcal{G}_t$ -measurable.

(h) Mapping onto the mixture of HMM regressions (Chamroukhi & Nguyen §4.2).

Chamroukhi & Nguyen	Our model
n curves	one long series ($n = 1$)
K clusters of curves	one cluster ($\tau_{i1} = 1$)
R_k regimes	our K regimes
time points $j = 1, \dots, m_i$	dates $t = 1, \dots, T$
scalar y_{ij}	the vector $(m_{t+1}, r_{1,t+1}, \dots, r_{N_t,t+1})$
covariate x_j (polynomial in time)	the design matrix X_t of stock characteristics
diagonal W_{ikr} of γ_{ijr}	$\gamma_{t+1}(k)$ repeated over the stocks of each date

C References and what each one gives

Textbook. C. M. Bishop (2006), *Pattern Recognition and Machine Learning*:

- §3.1: linear basis-function models;
- §4.2–4.3: where the softmax comes from, and IRLS;
- §5.1–5.3: MLPs and backpropagation;
- §9.2–9.4: mixtures, EM, and the lower bound;
- §13.1–13.2: Markov chains, HMMs, forward–backward and Baum–Welch;
- §14.5: mixtures of linear regressions and mixtures of experts.

Warm-up reference. F. Chamroukhi and H. D. Nguyen (2018), *Model-Based Clustering and Classification of Functional Data*, arXiv:1803.00276. EM for regression mixtures (§3), the softmax-gated RHLF (§4.3), and the mixture of HMM regressions (§4.2).

The mixture-of-experts canon.

- R. A. Jacobs, M. I. Jordan, S. J. Nowlan and G. E. Hinton (1991), Adaptive mixtures of local experts, *Neural Computation* 3(1), 79–87. Eq. 1.3 is the per-observation mixture negative log-likelihood; eq. 1.5 is the responsibility-weighted gradient.
- M. I. Jordan and R. A. Jacobs (1994), Hierarchical mixtures of experts and the EM algorithm, *Neural Computation* 6(2), 181–214. Indicator variables, the complete-data likelihood, and the E-step posterior h .
- A. S. Weigend, M. Mangeas and A. N. Srivastava (1995), Nonlinear gated experts for time series: discovering regimes and avoiding overfitting, *International Journal of Neural Systems* 6(4), 373–399. MLP experts (10 tanh units) and an MLP gate (20 units); eqs. 5, 9–10, 12 are the derivation of Section 4. Takeaways:
 - expert-specific variances matter for discovering regimes (footnote 12);
 - the variance can be shrunk towards a prior value (eq. 23);
 - the expert gradient h/σ^2 amounts to a weighted regression (eq. 24);
 - a scatter of g against h is a useful diagnostic;
 - start with more experts than needed;
 - seeds are unstable when regime variances differ by less than a factor of three.
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Regimes and finance.

- J. D. Hamilton (1989), *Econometrica* 57(2), 357–384. The Markov-switching model and its filter.

- S. Gu, B. Kelly and D. Xiu (2020), Empirical asset pricing via machine learning, *Review of Financial Studies* 33(5), 2223–2273. The NN1–NN5 architectures.
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Approximation and EM theory.

- G. Cybenko (1989), Approximation by superpositions of a sigmoidal function, *Mathematics of Control, Signals and Systems* 2, 303–314.
- K. Hornik, M. Stinchcombe and H. White (1989), Multilayer feedforward networks are universal approximators, *Neural Networks* 2, 359–366.
- A. P. Dempster, N. M. Laird and D. B. Rubin (1977), Maximum likelihood from incomplete data via the EM algorithm, *Journal of the Royal Statistical Society B* 39, 1–38.
- R. M. Neal and G. E. Hinton (1998), A view of the EM algorithm that justifies incremental, sparse, and other variants, in *Learning in Graphical Models*. The lower-bound view of EM and generalised EM.

Project documents.

- `Model_Derivation_BaseCase.md`: Hamilton filter, Kim smoother, Baum–Welch, experts.
- `Advisor_Questions.md`: Q6, Q8, Q14, Q16, Q18, Q20–Q24.
- `MoE_Formulation_Course_Plan.md`: the plan behind these notes.

D Per date versus per row: the theory

This appendix gives the theory behind the decision in Section 7.3 (Q24, decided 2026-10-05): the experts are trained on the per-row objective ℓ_{row} , not on the per-date likelihood ℓ_{date} . It proceeds one step at a time:

1. the two models and their graphs (D.1);
2. the likelihood each one implies (D.2);
3. the key identity: both models give every single stock the same distribution (D.3);
4. composite likelihood, and why ℓ_{row} is one for our model (D.4);
5. why it still estimates the right parameters (D.5) and what it gives up (D.6);
6. how the responsibilities behave under each objective (D.7);
7. what happens when A3 is false (D.8);
8. what this means for a frozen gate (D.9), and a summary (D.10).

Throughout, fix the gate weights $\pi_k(t)$ (frozen, $\mathcal{F}_t$ -measurable) and write

$$\mathcal{N}_{ik} := \mathcal{N}(r_{i,t+1}; \mu_k(x_{i,t}), s_k^2), \quad \theta = (\psi_1, \dots, \psi_K, s_1, \dots, s_K).$$

D.1 Two models

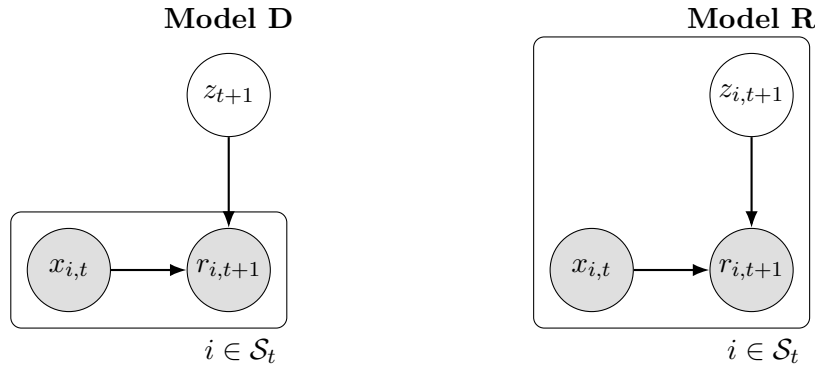
Model D (per date). This is the model of Section 6. On each date one regime is drawn and every stock uses it:

$$z_{t+1} \sim \text{Cat}(\pi_1(t), \dots, \pi_K(t)), \quad r_{i,t+1} \mid z_{t+1} = k \sim \mathcal{N}(\mu_k(x_{i,t}), s_k^2) \text{ independently over } i.$$

Model R (per row). Each stock-day draws its own regime from the same prior:

$$z_{i,t+1} \sim \text{Cat}(\pi_1(t), \dots, \pi_K(t)) \text{ independently over } i, \quad r_{i,t+1} \mid z_{i,t+1} = k \sim \mathcal{N}(\mu_k(x_{i,t}), s_k^2).$$

The only difference is where the latent node sits: outside the stock plate (D) or inside it (R).



D.2 The likelihood each model implies

Model D. Sum rule over the one latent of the date, then A3 (independence given the regime):

$$p_D(r_{\cdot,t+1}) = \sum_{k=1}^K \pi_k(t) \prod_{i \in \mathcal{S}_t} \mathcal{N}_{ik} \quad \Rightarrow \quad \ell_{\text{date}}(\theta) = \sum_t \log \sum_k \pi_k(t) \prod_i \mathcal{N}_{ik}. \quad (126)$$

Model R. Each stock has its own latent, so the sum over latents factorises stock by stock (the same step as Exercise A1(c)):

$$p_R(r_{\cdot,t+1}) = \prod_{i \in \mathcal{S}_t} \sum_{k=1}^K \pi_k(t) \mathcal{N}_{ik} \Rightarrow \ell_{\text{row}}(\theta) = \sum_t \sum_i \log \sum_k \pi_k(t) \mathcal{N}_{ik}. \quad (127)$$

In D the sum sits outside the product; in R it sits inside.

D.3 The key identity: identical one-stock distributions

Proposition 1

Under Model D, the distribution of one stock's return on its own is

$$p_D(r_{i,t+1} \mid x_{\cdot,t}, \mathcal{F}_t) = \sum_{k=1}^K \pi_k(t) \mathcal{N}_{ik},$$

which is exactly the factor of stock i in Model R. The two models agree on every single stock and differ only in how stocks move *together*.

Part 1: the independence used (d-separation). In Model D the only path between $r_{i,t+1}$ and $r_{j,t+1}$ runs through z_{t+1} . Conditioning on z_{t+1} blocks it (a tail-to-tail node), so given the regime the returns are independent. That is A3(a), read off the graph.

Part 2: the algebra. Start from the joint of all returns and integrate out every stock except i :

$$p_D(r_{i,t+1}) = \int \sum_k \pi_k(t) \prod_j \mathcal{N}_{jk} \prod_{j \neq i} dr_{j,t+1}.$$

The sum over k is finite, so it may be taken outside the integral:

$$= \sum_k \pi_k(t) \mathcal{N}_{ik} \prod_{j \neq i} \int \mathcal{N}_{jk} dr_{j,t+1}.$$

Every remaining integral is the integral of a density, which is one:

$$= \sum_k \pi_k(t) \mathcal{N}_{ik}. \quad \blacksquare$$

Consequences.

- The point forecast $\sum_k \pi_k \mu_k$ and the whole one-stock predictive distribution are the same in both models.
- The models differ in the covariance between stocks. Writing $\mu_{ik} = \mu_k(x_{i,t})$, under D (Section 8.5),

$$\text{Cov}_D(r_{i,t+1}, r_{j,t+1}) = \sum_k \pi_k(t) (\mu_{ik} - \bar{\mu}_i) (\mu_{jk} - \bar{\mu}_j), \quad \bar{\mu}_i = \sum_k \pi_k \mu_{ik},$$

while under R it is zero. Stocks co-move under D because they share an unknown regime.

D.4 Composite likelihood

Definition (Lindsay 1988). Split the data into pieces $A_1, \dots, A_M$ (single observations, pairs, blocks) whose densities are known under the model. The composite log-likelihood is the weighted sum of the piece log-likelihoods:

$$\ell_C(\theta) = \sum_{m=1}^M w_m \log p(y \in A_m; \theta).$$

It is a full likelihood only if the pieces happen to be independent. When each piece is one observation and the weights are one, ℓ_C is the *independence likelihood*: the log-likelihood one would write if the observations were independent, evaluated with their correct marginals (Varin, Reid & Firth 2011).

Application. By Proposition 1, the correct marginal of stock-day (i, t) under Model D is $\sum_k \pi_k \mathcal{N}_{ik}$. Hence

$$\ell_{\text{row}}(\theta) = \sum_t \sum_i \log p_D(r_{i,t+1}; \theta) \quad (128)$$

is the independence likelihood of Model D. The same function therefore has two readings: the exact likelihood of Model R, and a composite likelihood of Model D. Our study adopts the second reading. The model stays Model D; only the estimation objective changes.

D.5 Why it estimates the right parameters

Step 1: each piece has an unbiased score. Let θ_0 be the true parameter and $p_i(r; \theta)$ the marginal density of piece i . Differentiating the identity that a density integrates to one,

$$\int p_i(r; \theta) dr = 1 \quad \Rightarrow \quad \int \frac{\partial p_i(r; \theta)}{\partial \theta} dr = 0 \quad \Rightarrow \quad \mathbb{E}_{\theta_0} \left[\frac{\partial}{\partial \theta} \log p_i(r; \theta) \Big|_{\theta_0} \right] = 0.$$

The last step writes the derivative of p_i as p_i times the derivative of $\log p_i$ and evaluates at θ_0 , where p_i is the true density.

Step 2: a sum of unbiased scores is unbiased. Expectation is linear, and it does not care whether the pieces are dependent:

$$\mathbb{E}_{\theta_0} [\nabla \ell_{\text{row}}(\theta_0)] = \sum_t \sum_i \mathbb{E}_{\theta_0} [\nabla \log p_i(r_{i,t+1}; \theta_0)] = 0.$$

Dependence between the pieces would matter for the *variance* of the sum, never for its mean.

Step 3: consistency. The estimator solves $\nabla \ell_{\text{row}}(\hat{\theta}) = 0$, an unbiased estimating equation. Under the usual regularity conditions it is consistent: as the number of dates grows, $\hat{\theta} \rightarrow \theta_0$. The conditions are a law of large numbers across dates and identifiability of θ from the one-stock marginals. Identifiability holds under mild conditions: the marginal is a finite Gaussian mixture of regressions, which is identified up to relabelling of the experts, and the frozen gate removes the relabelling by tying expert k to HMM state k .

D.6 What it gives up: efficiency and standard errors

Define the two matrices of estimating-equation theory:

$$H(\theta_0) = -\mathbb{E}[\nabla^2 \ell_{\text{row}}], \quad J(\theta_0) = \text{Var}[\nabla \ell_{\text{row}}].$$

For a full likelihood $H = J$, the Fisher information. For a composite likelihood they differ, because the pieces are dependent, and the asymptotic variance is the *sandwich*

$$\text{Var}(\hat{\theta}) \approx H^{-1} J H^{-1} = G^{-1}, \quad G = H J^{-1} H \text{ (the Godambe information).}$$

Two consequences:

- **Efficiency.** G is never larger than the Fisher information of the full likelihood. The information lost is exactly what the co-movement of stocks on a date says about the regime.
- **Standard errors.** The inverse Hessian alone, H^{-1} , is not a valid variance. Overlapping 5-day targets add dependence across dates, which works the same way.

Neither matters for the study, because inference comes from walk-forward rank ICs with Hansen–Hodrick corrections, never from the likelihood’s curvature.

D.7 How the responsibilities behave

Bayes’ rule gives the posterior odds of regimes k and j after seeing the returns.

Per row: one stock’s evidence.

$$\log \frac{\gamma_{i,t}(k)}{\gamma_{i,t}(j)} = \log \frac{\pi_k(t)}{\pi_j(t)} + \log \frac{\mathcal{N}_{ik}}{\mathcal{N}_{ij}}.$$

Per date: the sum of all stocks’ evidence.

$$\log \frac{\gamma_t(k)}{\gamma_t(j)} = \log \frac{\pi_k(t)}{\pi_j(t)} + \sum_{i \in S_t} \log \frac{\mathcal{N}_{ik}}{\mathcal{N}_{ij}}.$$

Size of the evidence. If regime k is true, each log-ratio term has expectation $\text{KL}(\mathcal{N}_{\cdot k} \parallel \mathcal{N}_{\cdot j}) > 0$. With the example of Section 7.5 this is about 0.22 nats per stock.

- Per row the evidence is 0.22 nats, a likelihood ratio of about 1.25. The posterior stays close to the gate’s prior.
- Per date it is about $0.22 \times 500 = 110$ nats. The posterior is 0 or 1 whatever the gate said.

D.8 When A3 is false

A model in which A3 fails but the marginals are right. Let stocks belong to clusters c (industries) of size m , and add a shared cluster shock inside each regime:

$$r_{i,t+1} = \mu_k(x_{i,t}) + u_{c(i),t+1} + e_{i,t+1}, \quad u_c \sim \mathcal{N}(0, \rho s_k^2), \quad e_i \sim \mathcal{N}(0, (1 - \rho) s_k^2),$$

independently. Two facts follow.

1. **The one-stock marginal is unchanged.** The sum of two independent Gaussians is Gaussian with the variances added, $\rho s_k^2 + (1 - \rho)s_k^2 = s_k^2$, so each stock is still $\sum_k \pi_k \mathcal{N}_{ik}$. The per-row objective is *correctly specified*, and D.5 applies unchanged.
2. **The joint is no longer a product.** Two stocks in the same cluster are correlated at ρ within a regime. The per-date likelihood, which multiplies their densities, is *misspecified*.

How much evidence the per-date likelihood overcounts. Write $v_i = \log(\mathcal{N}_{ik}/\mathcal{N}_{ij})$ for stock i 's evidence term in D.7, with variance σ_v^2 , and let ρ_v be the correlation of two such terms in the same cluster. It equals the return correlation ρ when the two experts differ in their means, because v_i is then linear in the return, and ρ^2 when they differ only in their noise levels, because v_i is then linear in the squared residual. The variance of a sum of N terms, in clusters of m with correlation ρ_v inside a cluster and zero across clusters, is

$$\text{Var}\left(\sum_{i=1}^N v_i\right) = N\sigma_v^2(1 + (m-1)\rho_v),$$

which is the variance of N_{eff} independent terms, where

$$N_{\text{eff}} = \frac{N}{1 + (m-1)\rho_v}$$

(Kish's design effect). The per-date posterior treats the evidence as if it came from N independent stocks, so it is overconfident by the factor $N/N_{\text{eff}} = 1 + (m-1)\rho_v$. With $m = 45$ and $\rho_v = 0.1$ the factor is 5.4. In this illustration the values of m and ρ_v are assumed, not measured. In addition, the parameters that maximise a misspecified likelihood need not be θ_0 : they are the pseudo-true values that make the *wrong joint* fit best. When the regimes are far apart this does little harm, because every date is still assigned correctly. When they are close, which is the realistic case for experts that differ by small corrections, overconfident misassignments feed back into the experts' means and noise levels.

A simulation check. Simulate Model D with the cluster shock above: $K = 2$, $T = 600$ dates, $N = 200$ stocks in clusters of $m = 20$, gate weights drawn uniformly in $[0.1, 0.9]$, and two close regimes with means 0 and 0.05 and noise levels 1 and 1.08. Then fit the means and noise levels by maximising each objective.

	μ_1	μ_2	s_1	s_2
True values	0	0.05	1	1.08
A3 holds ($\rho = 0$): per row	-0.003	0.050	0.994	1.088
A3 holds ($\rho = 0$): per date	-0.002	0.049	1.001	1.082
A3 fails ($\rho = 0.5$): per row	-0.017	0.070	0.995	1.079
A3 fails ($\rho = 0.5$): per date	-0.150	0.230	1.017	1.023

When A3 holds, both objectives recover the truth. When it fails, the per-row estimates stay close to it, while the per-date estimates are pulled away: the gap between the means is nearly eight times too large (0.38 against 0.05), and the noise levels are squeezed together. Even at the true parameters, the per-date posterior puts more than half its weight on the wrong regime on 14% of dates when $\rho = 0$ and on 35% when $\rho = 0.5$. This is one seed of a stylised example, not a measurement on the data.

The general principle. An objective built from a lower-dimensional piece of the model (here, single stocks) needs only that piece to be right. This is the logic of generalised estimating equations with a working independence assumption (Liang & Zeger 1986). The per-date likelihood needs the whole joint distribution to be right, and in a cross-section of 500 stocks it is not.

D.9 What this means for a frozen gate

- The gradient for expert k on row (i, t) is the regression residual weighted by the responsibility (Section 7.4). The responsibility therefore decides which data each expert learns from.
- **Per date:** $\gamma_t(k) \in \{0, 1\}$ is set by the day's returns (D.7) and inflated by correlated evidence (D.8). Each expert learns from the days whose cross-section fits it, and the cross-section re-decides the regime every day. This is the daily version of the joint fit that Section 7.2 rules out, because it would let the stocks redefine the regimes.
- **Per row:** $\gamma_{i,t}(k)$ stays near $\pi_k(t)$. Each expert learns mainly from the days the gate assigns to its regime. The regimes remain *market* regimes, which is why the gate was frozen (Q19).
- **The cost:** specialisation is softer, and the expert with the larger s_k can attract outlying rows. Diagnostic: compare each expert's training weight with $\pi_k(t)$ and with the size of the residuals.

D.10 Summary

	ℓ_{date}	ℓ_{row} (used)
Status under Model D	exact likelihood	independence (composite) likelihood
One-stock distribution	$\sum_k \pi_k \mathcal{N}_{ik}$	identical (Proposition 1)
Needs A3(a)	yes, the full joint	no, only the marginals
Consistent if A3 fails	not guaranteed; distorted when regimes are close (D.8)	yes (D.5, D.8)
Efficiency if the model is exactly right	full	lower (Godambe, D.6)
Standard errors from the Hessian	valid if A3 holds	invalid; sandwich needed
Responsibilities	0 or 1, set by the returns	close to the gate's π
Who defines the regime in training	the cross-section	the frozen gate
Mini-batches	whole dates	any rows

References: Lindsay (1988), Contemporary Mathematics 80; Varin, Reid & Firth (2011), Statistica Sinica 21; Liang & Zeger (1986), Biometrika 73(1); Pakel, Shephard, Sheppard & Engle (2021), JBES 39(3); Kish (1965), *Survey Sampling*; Godambe (1960), Annals of Mathematical Statistics 31.